Analysis I

Scribed Course Notes

Definitions
Results
Examples
Notes

Contents

1 Foundations and Number Systems **1**
1.1 Natural Numbers and Induction . . . 1
1.2 Integers and Rational Numbers . . . 4
1.3 Sets, Maps, and Countability . . . 6

2 The Real Numbers **10**
2.1 Construction from Cauchy Sequences . . . 10
2.2 The Least-Upper-Bound Property . . . 17
2.3 Powers and Roots . . . 21
2.4 The Nested-Interval Property . . . 22
2.5 Uncountability and Interval Bijections . . . 23
2.6 Cantor–Schröder–Bernstein . . . 24

3 Sequences and Series **26**
3.1 Limits of Sequences . . . 26
3.2 Euler's Number e . . . 31
3.3 Bolzano–Weierstrass . . . 32
3.4 Proper Divergence . . . 35
3.5 Series . . . 36
3.6 Convergence Tests . . . 41

4 Topology and Compactness **44**
4.1 Topological Spaces, Bases, and Separability . . . 44
4.2 Closed Sets and Compactness . . . 47
4.3 Cantor Set . . . 51

5 Limits and Continuity **54**
5.1 Limits of Functions . . . 54
5.2 Continuous Functions . . . 56
5.3 Extension Theorems . . . 58
5.4 Continuity over Compact Sets . . . 59
5.5 Uniform Continuity . . . 61
5.6 Types of Discontinuities . . . 63
5.7 Monotone and Inverse Functions . . . 63
5.8 Lipschitz and Hölder Continuity . . . 68

6 Differentiation **71**
6.1 The Derivative and Inverse Functions . . . 71
6.2 Mean Value Theorem . . . 73
6.3 L'Hôpital's Rule . . . 78
6.4 Higher-Order Derivatives and Taylor's Theorem . . . 81

7 Convexity **85**

1 Foundations and Number Systems

1.1 Natural Numbers and Induction

1.1.1 Peano Axioms

Peano Axioms for $\mathbb{N}$

A1. $0 \in \mathbb{N}$.

A2. For every $n \in \mathbb{N}$, there exists a successor n^+, where $n^+ = n + 1$.

A3. 0 is not a successor of any $n \in \mathbb{N}$.

A4. If $x, y \in \mathbb{N}$ and $x^+ = y^+$, then $x = y$.

A5. Mathematical induction:

$$\left.\begin{array}{l} P(0) \text{ is true,} \\ P(n) \text{ true } \Longrightarrow\ P(n^+) \text{ true} \end{array}\right\} \Longrightarrow\ P(n) \text{ holds for every } n \in \mathbb{N}.$$

Definition

For $m \in \mathbb{N}$, addition is defined by

$$0 + m = m, \qquad n^+ + m = (n + m)^+.$$

Fact 1

For all $m, n \in \mathbb{N}$,

$$n + m^+ = (n + m)^+.$$

Proof

For $n = 0$,

$$0 + m^+ \overset{\text{definition}}{=} m^+ \overset{\text{definition}}{=} (0 + m)^+ \qquad \text{(base case)}.$$

For the induction step, suppose $P(n)$ is true. Then

$$n^+ + m^+ \overset{\text{definition}}{=} (n + m^+)^+ \overset{\text{induction hypothesis}}{=} ((n + m)^+)^+ \overset{\text{definition}}{=} (n^+ + m)^+.$$

□

Fact 2 (Commutativity)

$$n + m = m + n.$$

The proof uses induction.

Fact 3 (Associativity)

$$(n + m) + \ell = n + (m + \ell).$$

Fact 4 (Cancellation law)

$$m + \ell = n + \ell \implies m = n.$$

Definition

$n \in \mathbb{N}$ is *positive* if and only if $n \neq 0$.

Fact

For $n, m \in \mathbb{N}$, if n is positive, then $n + m$ is positive (by mathematical induction).

Definition

For $m, n \in \mathbb{N}$,

$$\begin{aligned} m \geq n &\iff \text{there exists } \ell \in \mathbb{N} \text{ such that } n + \ell = m, \\ m > n &\iff \text{there exists } \ell \in \mathbb{N} \text{ such that } n + \ell^+ = m, \\ m \leq n &\iff \text{there exists } \ell \in \mathbb{N} \text{ such that } m + \ell = n, \\ m < n &\iff \text{there exists } \ell \in \mathbb{N} \text{ such that } m + \ell^+ = n. \end{aligned}$$

Fact (Reflexivity)

$$\ell \geq \ell \qquad (\ell \leq \ell).$$

Fact (Transitivity)

$$\ell \geq m, \quad m \geq n \implies \ell \geq n.$$

Fact

$$m \geq \ell, \quad \ell \geq m \implies m = \ell.$$

Fact

$$\ell \geq m \iff \ell + n \geq m + n, \qquad \forall n \in \mathbb{N}.$$

Fact

$$\ell < m \implies \ell^+ \leq m.$$

Fact

$$\ell < m \implies \exists q \in \mathbb{N}_{>0} \text{ such that } \ell + q = m.$$

Fact (Trichotomy)

For every $m, n \in \mathbb{N}$, either $m > n$, $m < n$, or $m = n$.

Definition

Multiplication is defined, for $m, n \in \mathbb{N}$, by

$$0 \times m = 0, \qquad n^+ \times m = (n \times m) + m \quad \text{inductively.}$$

Facts

1. Commutativity.
2. Distributivity.
3. Associativity.
4. If $m < n$ and ℓ is positive, then $m\ell < n\ell$.
5. If $\ell \times n = m \times n$ and n is positive, then $\ell = m$.
6. For every $q \in \mathbb{N}_{>0}$ and $m \in \mathbb{N}$, there exist unique $n, r \in \mathbb{N}$ such that
$$m = nq + r, \qquad r < q.$$
7. If $n \in \mathbb{N}$ and $n > 1$, then n can be uniquely written as a product of primes (Fundamental Theorem of Arithmetic).

Proof of Fact 6

Fix $q \in \mathbb{N}_{>0}$. We prove existence by induction on m. For $m = 0$, take $n = r = 0$. Suppose

$$m = nq + r, \qquad 0 \le r < q.$$

Since $r + 1 \le q$, either $r + 1 < q$, in which case

$$m + 1 = nq + (r + 1),$$

or $r + 1 = q$, in which case

$$m + 1 = (n + 1)q.$$

For uniqueness, suppose

$$nq + r = n'q + r', \qquad 0 \le r, r' < q.$$

If $n > n'$, then $(n - n')q = r' - r$, whose left-hand side is at least q while its right-hand side is less than q, a contradiction. Interchanging the two representations rules out $n' < n$. Thus $n = n'$, and then $r = r'$. □

1.1.2 Strong Mathematical Induction

Theorem (Strong Mathematical Induction)

Let $m_0 \in \mathbb{N}$. Suppose $P(m_0)$ is true and, for every $m \geq m_0$,

$$[\forall n \in \mathbb{N}, \quad m_0 \leq n \leq m \implies P(n)] \implies P(m+1).$$

Then $P(m)$ is true for every $m \geq m_0$.

Discrete Cauchy–Schwarz Inequality

For $a_1, \ldots, a_N, b_1, \ldots, b_N \in \mathbb{R}$,

$$\left(\sum_{i=1}^{N} a_i b_i\right)^2 \leq \left(\sum_{i=1}^{N} a_i^2\right)\left(\sum_{i=1}^{N} b_i^2\right).$$

Proof

For every $t \in \mathbb{R}$,

$$\sum_{i=1}^{N}(a_i t + b_i)^2 = t^2 \sum_{i=1}^{N} a_i^2 + 2t \sum_{i=1}^{N} a_i b_i + \sum_{i=1}^{N} b_i^2 \geq 0.$$

If $\sum_i a_i^2 = 0$, then every $a_i = 0$ and the result is immediate. Otherwise this quadratic in t has nonpositive discriminant, so

$$4\left(\sum_{i=1}^{N} a_i b_i\right)^2 - 4\left(\sum_{i=1}^{N} a_i^2\right)\left(\sum_{i=1}^{N} b_i^2\right) \leq 0.$$

Rearranging proves the inequality. □

1.2 Integers and Rational Numbers

1.2.1 Subtraction

Definition

Define subtraction on $\mathbb{N}$ by

$$m_1 -_{\mathbb{N}} n_1 = m_2 -_{\mathbb{N}} n_2 \iff m_1 + n_2 = m_2 + n_1, \qquad \forall m_1, m_2, n_1, n_2 \in \mathbb{N}.$$

1.2.2 The Integers $\mathbb{Z}$

Definition

$$\mathbb{Z} := \{m -_{\mathbb{N}} n \mid m, n \in \mathbb{N}\}, \qquad \text{the set of integers.}$$

Denote

$$-n := 0 -_{\mathbb{N}} n.$$

Definition

An order on $\mathbb{Z}$ is given by

$$\underbrace{m_1 -_{\mathbb{N}} n_1}_{z_1} \geq \underbrace{m_2 -_{\mathbb{N}} n_2}_{z_2} \iff m_1 + n_2 \geq m_2 + n_1.$$

Definition

Subtraction on $\mathbb{Z}$ is defined by

$$\begin{aligned} a - b &= (m_1 -_{\mathbb{N}} n_1) - (m_2 -_{\mathbb{N}} n_2) \\ &:= (m_1 + n_2) -_{\mathbb{N}} (m_2 + n_1). \end{aligned}$$

Check

$$\underset{\text{subtraction on } \mathbb{Z}}{\mathbb{Z}} \text{ is equivalent as } \underset{\text{subtraction on } \mathbb{N}}{\mathbb{N}}.$$

Note

We can extend the definition of multiplication on $\mathbb{N}$ to $\mathbb{Z}$ by defining

$$(-1) \times (-1) = 1.$$

Theorem (Fundamental theorem of arithmetic)

Every $m \in \mathbb{N}$ with $m > 1$ can be written, uniquely up to the order of the factors, as a product of primes. Equivalently,

$$m = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k},$$

where the p_i are distinct primes and $\alpha_i \in \mathbb{N}_{>0}$; the primes and exponents are unique up to reordering.

Proof

For existence, use strong induction on m. If m is prime, it is already a product of primes. If it is composite, write $m = ab$ with $1 < a, b < m$ and apply the induction hypothesis to both a and b.
For uniqueness, suppose

$$p_1 p_2 \cdots p_t = q_1 q_2 \cdots q_s,$$

where all displayed factors are prime, with repetitions allowed. By Euclid's lemma, p_1 divides some q_j. Since q_j is prime, $p_1 = q_j$. Reorder the right-hand side so that $q_j = q_1$, cancel this common factor, and repeat. Induction gives $s = t$ and equality of the two lists up to reordering. Grouping repeated primes yields the stated exponent form. □

Definition

$$\mathbb{Q} := \left\{ \frac{p}{q} \;\middle|\; p \in \mathbb{Z},\ q \in \mathbb{Z} \setminus \{0\} \right\}.$$

Theorem

If $x^2 = 2$, then $x \notin \mathbb{Q}$.

Proof

Suppose there exist $p, q \in \mathbb{Z}$ with $(p, q) = 1$ such that $x = p/q$. Then

$$\begin{aligned}\left(\frac{p}{q}\right)^2 = 2 &\implies p^2 = 2q^2 \implies 2 \mid p \implies p = 2p_1 \\ &\implies 4p_1^2 = 2q^2 \implies 2p_1^2 = q^2 \implies 2 \mid q \\ &\implies 2 \mid (p, q),\end{aligned}$$

contradicting $(p, q) = 1$. □

1.3 Sets, Maps, and Countability

Theorem

$$\left(\bigcup_{n=1}^{\infty} A_n\right)^c = \bigcap_{n=1}^{\infty} A_n^c, \qquad \left(\bigcap_{n=1}^{\infty} A_n\right)^c = \bigcup_{n=1}^{\infty} A_n^c.$$

Properties

$$f(C_1 \cap C_2) \subset f(C_1) \cap f(C_2), \qquad f(C_1 \cup C_2) = f(C_1) \cup f(C_2).$$

Properties

$$\begin{aligned}f^{-1}(D_1 \cap D_2) &= f^{-1}(D_1) \cap f^{-1}(D_2), \\ f^{-1}(D_1 \cup D_2) &= f^{-1}(D_1) \cup f^{-1}(D_2), \\ f^{-1}(D^c) &= (f^{-1}(D))^c.\end{aligned}$$

Note

The identity

$$f(D^c) = (f(D))^c$$

is not always true. Rather,

$$(f(D))^c \subset f(D^c) \quad \text{is true if } f \text{ is surjective,}$$

with equality if f is bijective, and no inclusion if f is neither.

Example

Let

$$f : \mathbb{Z} \longrightarrow \mathbb{N}, \qquad f(x) = x^2.$$

Then

$$\text{Range } f \subsetneq \mathbb{N}, \qquad 2 \notin \text{Range } f.$$

Definition

$$\begin{aligned} f \text{ is injective}: &\quad f(x_1) = f(x_2) \implies x_1 = x_2, \\ f : A \to B \text{ is surjective}: &\quad \forall y \in B,\ \exists x \in A \text{ such that } f(x) = y, \\ f \text{ is bijective}: &\quad f \text{ is injective and surjective.} \end{aligned}$$

Terminology

Simple function, step function, characteristic function.

Example

For $f(x) = x + 1$, the map $f : \mathbb{Z} \to \mathbb{Z}$ is bijective. The map $f : \mathbb{N} \to \mathbb{N}$ is injective but not surjective.

Proposition

$$f^{-1}\left(\bigcup_{n=1}^{\infty} D_n\right) = \bigcup_{n=1}^{\infty} f^{-1}(D_n), \qquad f^{-1}\left(\bigcap_{n=1}^{\infty} D_n\right) = \bigcap_{n=1}^{\infty} f^{-1}(D_n).$$

Proof

$$\begin{aligned} x \in f^{-1}\left(\bigcup_{n=1}^{\infty} D_n\right) &\iff f(x) \in \bigcup_{n=1}^{\infty} D_n \\ &\iff \exists D_n \text{ such that } f(x) \in D_n \\ &\iff x \in f^{-1}(D_n) \\ &\iff x \in \bigcup_{n=1}^{\infty} f^{-1}(D_n). \end{aligned}$$

□

Definition

Sets A, B have the same cardinality if there exists a bijection $f : A \to B$. Denote this by $A \sim B$.
A finite set has cardinality equal to the number of elements in it.
A is countable if $A \sim \mathbb{N}$.
A is uncountable if A is neither finite nor countable.

Example

$|\mathbb{Z}| = |\mathbb{N}|$. One such bijection is illustrated below:

$$\begin{array}{ccccccccccc} \cdots & & -3 & -2 & -1 & 0 & 1 & 2 & 3 & \cdots \\ & & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \downarrow & \\ \cdots & & 5 & 3 & 1 & 0 & 2 & 4 & 6 & \cdots \end{array}$$

Proposition

Let $A \subset B$.

- If B is finite or countable, then A is finite or countable.
- If A is uncountable, then B is uncountable.

Lemma

Every $S \subset \mathbb{N}$ is either finite or countable.

Proof

If S is finite, the conclusion is immediate. If S is infinite, by the well-ordering principle on $\mathbb{N}$ there exists a least element of S. If we keep taking out the least element of S, we get a sequence of $\mathbb{N}$ which we can enumerate by $\mathbb{N}$. □

Definition

The Cartesian product is

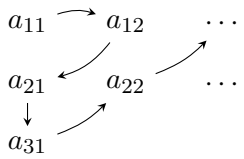
$$A \times B := \{(a, b) \mid a \in A,\ b \in B\}.$$

Proposition

A countable union of countable sets is countable.

Proof

Enumerate the array diagonally:



The indices form an infinite subset of $\mathbb{N}$, so the union is countable. □

Proposition

If A, B are countable, then $A \times B$ is countable.

Proof

Choose bijections $\alpha : A \to \mathbb{N}$ and $\beta : B \to \mathbb{N}$. Then

$$(a, b) \longmapsto (\alpha(a), \beta(b))$$

is a bijection from $A \times B$ to $\mathbb{N} \times \mathbb{N}$, and the latter is countable by diagonal enumeration. □

Proposition

If $A_1, \dots, A_n$ are countable, then

$$\prod_{i=1}^{n} A_i$$

is countable.

Proof

By induction. □

Proposition

Let $A \neq \varnothing$. The following are equivalent:

1. A is finite or countable.
2. There exists an injective map $f : A \to \mathbb{N}$.
3. There exists a surjective map $f : \mathbb{N} \to A$.

Proof

$$(1) \implies (3) \implies (2) \implies (1).$$

□

Proposition

$\mathbb{Q}$ is countable.

Proof

The map

$$\pi : \mathbb{Z} \times (\mathbb{Z} \setminus \{0\}) \longrightarrow \mathbb{Q}, \qquad \pi(p, q) = \frac{p}{q},$$

is surjective. Its domain is countable, so $\mathbb{Q}$ is finite or countable. Since $\mathbb{Z} \subset \mathbb{Q}$, it is infinite and therefore countable. □

Proof (alternative proof that a countable union of countable sets is countable)

We use the fundamental theorem of arithmetic. Let

$$S := \{2, 3, 5, 7, \dots\}, \qquad |S| = |\mathbb{N}|.$$

Define

$$B_n := \{p_n^k \mid k \in \mathbb{N}_{>0}\}, \qquad n \geq 1,$$

where $p_1 = 2, p_2 = 3, p_3 = 5, \dots$ enumerate the primes. Thus

$$B_1 = \{2, 4, 8, 16, \dots\}, \dots, \qquad |B_n| = |\mathbb{N}|, \quad n \geq 1.$$

Suppose the A_n are finite or countable. Discard any empty A_n, since they do not affect the

union. For every remaining n there exists a surjection

$$f_n : B_n \longrightarrow A_n.$$

Let

$$B = \bigcup_{n=1}^{\infty} B_n.$$

For $n \neq m$, the sets B_n and B_m are disjoint by unique prime factorization. Thus define

$$f|_{B_n} = f_n.$$

Then $f : B \to \bigcup A_n$ is surjective, by surjectivity of the f_n. The set B is an infinite subset of $\mathbb{N}$, hence it is countable. Therefore $\bigcup_n A_n$, as the image of B under a surjection, is finite or countable. □

2 Construction and Completeness of $\mathbb{R}$

2.1 Construction from Cauchy Sequences

Using equivalence classes of Cauchy sequences.

Proposition (Triangle Inequality)

For every $r_1, r_2 \in \mathbb{Q}$,

$$|r_1 + r_2| \leq |r_1| + |r_2|.$$

Proof

Notice that

$$-|r_1| - |r_2| \leq r_1 + r_2 \leq |r_1| + |r_2|.$$

□

Definition

We say

$$\lim_{n\to\infty} r_n = r$$

if, for every $\varepsilon > 0$ with $\varepsilon \in \mathbb{Q}$, there exists $N \in \mathbb{N}$ such that, for every $n \geq N$,

$$|r_n - r| < \varepsilon.$$

Definition

A sequence is Cauchy if, for every $\varepsilon > 0$ with $\varepsilon \in \mathbb{Q}$, there exists $N \in \mathbb{N}$ such that, for every $n, m \geq N$,

$$|r_n - r_m| < \varepsilon.$$

Lemma

Let $\{a_n\}_n \subset \mathbb{Q}$. If $\{a_n\}_n$ converges in $\mathbb{Q}$, then $\{a_n\}_n$ is Cauchy.

Proof

Use the triangle inequality. □

Example

The sequence $\{2^{-n}\}_n$ is Cauchy.

Proof

Choose

$$N \geq \log_2 \frac{1}{\varepsilon}.$$

Then

$$\left|\frac{1}{2^n} - \frac{1}{2^\ell}\right| < \varepsilon.$$

□

Definition

A sequence $\{a_n\}_{n=1}^{\infty}$ is bounded if there exists $M \in \mathbb{Q}^+$ such that

$$|a_n| \leq M, \qquad \forall n \in \mathbb{N}.$$

Lemma

If $\{a_n\}_n$ is Cauchy, then $\{a_n\}_n$ is bounded.

Proof

Trivial. □

Lemma

If $\{a_n\}_n$ and $\{b_n\}_n$ are Cauchy, then $\{a_n + b_n\}_n$ and $\{a_n b_n\}_n$ are Cauchy.

Proof

The assertion for $a_n + b_n$ follows directly from the triangle inequality. For the product, choose $M_a, M_b > 0$ such that $|a_n| \leq M_a$ and $|b_n| \leq M_b$ for every n. Given $\varepsilon > 0$, put

$$\delta := \frac{\varepsilon}{2(M_a + M_b)}.$$

For all sufficiently large n, ℓ, both $|a_n - a_\ell| < \delta$ and $|b_n - b_\ell| < \delta$. Hence

$$\begin{aligned} |a_n b_n - a_\ell b_\ell| &\leq |a_n b_n - a_n b_\ell| + |a_n b_\ell - a_\ell b_\ell| \\ &\leq M_a |b_n - b_\ell| + M_b |a_n - a_\ell| \\ &< (M_a + M_b)\delta < \varepsilon. \end{aligned}$$

□

Definition

Two sequences $\{a_n\}_n$ and $\{b_n\}_n$ are equivalent if, for every $\varepsilon > 0$, there exists $N > 0$ such that

$$|a_n - b_n| < \varepsilon, \qquad \forall n \geq N.$$

Example

The sequences $\{2^{-n}\}_n$ and $\{(-\frac{1}{2})^n\}_n$ are equivalent.

Key Definition (The reals $\mathbb{R}$)

Let

$$\mathcal{C} := \left\{ \{a_n\}_n \;\middle|\; \{a_n\}_n \text{ is a Cauchy sequence in } \mathbb{Q} \right\}.$$

The real numbers are the equivalence classes

$$\mathbb{R} := \mathcal{C}/\sim .$$

Write $[a_n]$ for the class represented by $\{a_n\}$. For $x = [a_n]$ and $y = [b_n]$, define

$$x + y := [a_n + b_n], \qquad xy := [a_n b_n].$$

These definitions do not depend on the chosen representatives. The zero element is

$$0_{\mathbb{R}} := [0].$$

A real number is nonzero when its class is not $[0]$.

Definition

Let $\{a_n\}_{n=1}^{\infty} \subset \mathbb{Q}$. It is

$$\begin{aligned}
\text{bounded away from 0} &\quad \text{if } \exists c > 0,\ c \in \mathbb{Q},\ \exists N \in \mathbb{N} \text{ such that } |a_n| \geq c,\ \forall n \geq N,\\
\text{bounded positively away from 0} &\quad \text{if } \exists c \in \mathbb{Q},\ c > 0,\ \exists N \in \mathbb{N} \text{ such that } a_n \geq c,\ \forall n \geq N,\\
\text{bounded negatively away from 0} &\quad \text{if } \exists c \in \mathbb{Q},\ c > 0,\ \exists N \in \mathbb{N} \text{ such that } a_n \leq -c,\ \forall n \geq N.
\end{aligned}$$

Definition

A real $x = [a_n] \in \mathbb{R}$ is positive if one representative $\{a_n\}$ is positively bounded away from 0, and negative if one representative is negatively bounded away from 0. Equivalence of Cauchy sequences shows that these definitions are independent of the representative.

Lemma

If $x \in \mathbb{R} \setminus \{0\}$, then x has a representative $\{a_n\} \subset \mathbb{Q}$ that is positively or negatively bounded away from 0.

Proof

Write $x = [a_n]$. Since $x \neq [0]$, the sequence $\{a_n\}$ does not converge to 0. Hence there is $\varepsilon_0 > 0$ such that arbitrarily far out one can find n with $|a_n| \geq \varepsilon_0$. Choose N so that

$$|a_m - a_n| < \frac{\varepsilon_0}{2}, \qquad m, n \geq N.$$

Choose $n \geq N$ with $|a_n| \geq \varepsilon_0$. If $a_n \geq \varepsilon_0$, then $a_m \geq \varepsilon_0/2$ for every $m \geq N$; if $a_n \leq -\varepsilon_0$, then $a_m \leq -\varepsilon_0/2$ for every $m \geq N$. □

Corollary

Every nonzero real number has a reciprocal.

Definition

Let $x \in \mathbb{R} \setminus \{0\}$. Choose a representative $\{a_n\}$ that is bounded away from 0 and, after changing finitely many initial terms if necessary, satisfies $a_n \neq 0$ for every n. Define

$$x^{-1} := \left[\frac{1}{a_n}\right].$$

The sequence $\{1/a_n\}$ is Cauchy, and the resulting class is independent of all choices.

Proposition (Trichotomy)

A real x is either zero, positive, or negative.

Proposition

For every $x \in \mathbb{R}$,

$$x \text{ is positive} \iff -x \text{ is negative} \iff \frac{1}{x} \text{ is positive if } x \neq 0.$$

Proposition

If $x, y > 0$, then

$$x + y > 0, \qquad xy > 0.$$

There are other analogous ones.

Definition

For every $x \in \mathbb{R}$,

$$|x| := \begin{cases} 0, & \text{if } x = 0, \\ x, & \text{if } x > 0, \\ -x, & \text{if } x < 0. \end{cases}$$

Definition

Define an order on $\mathbb{R}$ by

$$x > y \quad \text{if } x - y > 0, \qquad x < y \quad \text{if } x - y < 0.$$

Order and completeness

The order is defined directly from the Cauchy-sequence construction. The Archimedean property and density of $\mathbb{Q}$ follow from this definition; the least-upper-bound property is established below.

Theorem

1. Let $x \in \mathbb{R}$. Then there exist $r_1, r_2 \in \mathbb{Q}$ such that
$$r_1 < x < r_2.$$
2. (Archimedean property) For every $x \in \mathbb{R}$ and every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that
$$N\varepsilon > x.$$
3. (Density of $\mathbb{Q}$ in $\mathbb{R}$) For every $x, y \in \mathbb{R}$ with $x < y$, there exists $r \in \mathbb{Q}$ such that
$$x < r < y.$$

   A general, equivalent notion of density, for example in L^p, ℓ^p, and a normed vector space, is
$$\forall \varepsilon > 0,\ \forall x \in \mathbb{R},\ \exists r \in \mathbb{Q} \quad \text{such that} \quad \|x - r\| < \varepsilon.$$

Proof

1. If $x = 0$, pick $r_1 < 0$ and $r_2 > 0$. We show the result for $x > 0$; the case $x < 0$ follows by taking negatives. Suppose $x = [a_n] > 0$. By the preceding lemma, there are $c \in \mathbb{Q}_{>0}$ and $N_0 \in \mathbb{N}$ such that $a_n \geq c$ for $n \geq N_0$. Set $r_1 = c/2$. Then
$$[a_n - r_1] > 0, \qquad \text{so} \qquad x > r_1.$$
   Since $\{a_n\}$ is Cauchy, it is bounded above by some rational r_2; increasing r_2 if necessary gives $x < r_2$. Therefore $r_1 < x < r_2$.
2. There exists $r_1 \in \mathbb{Q}^+$ such that $\varepsilon > r_1$. There exists $r_2 \in \mathbb{Q}^+$ such that $r_2 > x$. Choose $N \in \mathbb{N}$ such that
$$Nr_1 > r_2 \qquad \text{(doable by the definition of } \mathbb{N}, \mathbb{Q}).$$
   Then
$$N\varepsilon > Nr_1 > r_2 > x.$$
   If $x \leq 0$, then it is trivial.

   If we knew the least-upper-bound property of $\mathbb{R}$: suppose for every n that $n \leq x/\varepsilon$. Then x/ε is an upper bound of $\mathbb{N}$. By the least-upper-bound property,

let $m = \sup \mathbb{N}$. Then $m - 1$ is not an upper bound, so there exists $n \in \mathbb{N}$ with $n > m - 1$. Hence $n + 1 \in \mathbb{N}$ and $n + 1 > m$, contradicting that m is an upper bound.

3. By the Archimedean property, there exists $n \in \mathbb{N}$ such that

$$n(y - x) > 1.$$

Let

$$S := \{s \in \mathbb{N} \mid s > nx\}.$$

Then $S \neq \varnothing$ by the Archimedean property. By the well-ordering principle on $\mathbb{N}$, S has a least element m. We claim that $m/n = r$ works. Since it is the least element,

$$(m - 1) \leq nx.$$

Since $n(y - x) > 1$, we have $ny > m$. Therefore

$$nx < m < ny \implies x < \frac{m}{n} < y.$$

This proves the result for $0 < x < y$. The case $x < 0 < y$ is trivial, and $x < y < 0$ follows by taking the negative.

Proof 2

Suppose

$$x = [a_n], \qquad y = [b_n].$$

Then

$$z = y - x > 0 \text{ in } \mathbb{R} \implies \exists r_0 \in \mathbb{Q}, r_0 > 0, \text{ such that } y - x > r_0.$$

Thus $b_n - a_n > r_0/2$ for all sufficiently large n. Choose $N \in \mathbb{N}$ large enough that this holds and, for every $n, \ell \geq N$,

$$|a_n - a_\ell| < \frac{r_0}{10}, \qquad |b_n - b_\ell| < \frac{r_0}{10}.$$

Let

$$r := a_N + \frac{r_0}{4}.$$

Then

$$a_n \leq a_N + \frac{r_0}{10} \leq a_N + \frac{r_0}{4} = r < a_n + \frac{r_0}{2} < b_n. \qquad (*)$$

Define

$$\{\widetilde{a}_n\} = \underbrace{a_N, a_N, \ldots, a_N}_{\text{first } N \text{ terms}}, a_{N+1}, a_{N+2}, \ldots,$$
$$\{\widetilde{b}_n\} = \underbrace{b_N, b_N, \ldots, b_N}_{N \text{ terms}}, b_{N+1}, b_{N+2}, \ldots.$$

Then

$$[\widetilde{a}_n] = [a_n], \qquad [\widetilde{b}_n] = [b_n].$$

Thus $[\widetilde{a}_n] = x$ and $[\widetilde{b}_n] = y$, so

$$x < r < y$$

by substituting into $(*)$.

2.1.1 Infinite Descent and Irrational Square Roots

Theorem

For every $k \in \mathbb{N}$ that is not a perfect square,

$$\sqrt{k} \notin \mathbb{Q}.$$

Proof

Suppose

$$\sqrt{k} = \frac{m}{n}, \qquad m, n \in \mathbb{N}_{>0}.$$

Then there exists $p \in \mathbb{N}$ such that

$$p < \sqrt{k} < p + 1,$$

where $p = \lfloor \sqrt{k} \rfloor$. Then

$$\begin{aligned} \sqrt{k} = \frac{m}{n} &= \frac{m(\sqrt{k} - p)}{n(\sqrt{k} - p)} \\ &= \frac{n\sqrt{k}(\sqrt{k} - p)}{n(\frac{m}{n} - p)} \\ &= \frac{nk - mp}{m - np} = \frac{p_1}{q_1}. \end{aligned}$$

Since $0 < \sqrt{k} - p < 1$,

$$\begin{aligned} p_1 &:= m(\sqrt{k} - p) = nk - mp \in \mathbb{N}, & 0 < p_1 < m, \\ q_1 &:= n(\sqrt{k} - p) = m - np \in \mathbb{N}, & 0 < q_1 < n. \end{aligned}$$

The same construction applied to p_1/q_1 gives a new representation with both numerator and denominator smaller again. Iterating would produce strictly decreasing sequences

$$m > p_1 > p_2 > \cdots, \qquad n > q_1 > q_2 > \cdots$$

in $\mathbb{N}$, contradicting the well-ordering principle. □

2.2 The Least-Upper-Bound Property

Definition

$$\sup A = \text{the least upper bound of } A, \qquad \inf A = \text{the greatest lower bound of } A.$$

Theorem (Least-Upper-Bound Property of $\mathbb{R}$)

Let $A \subset \mathbb{R}$. If $A \neq \varnothing$ and A has an upper bound, then A has a least upper bound.

Definition

If $A \neq \varnothing$ and A has no upper bound, define

$$\sup A := \infty.$$

If $A = \varnothing$, define

$$\sup A := -\infty.$$

Corollary

Let $A \subset \mathbb{R}$, $A \neq \varnothing$. If A is bounded below, then A has a greatest lower bound.

Proposition

$\mathbb{R} \setminus \mathbb{Q}$ (the irrationals) is dense in $\mathbb{R}$.

Proof

We will use $c = \sqrt{2} \in \mathbb{R} \setminus \mathbb{Q}$; assume this for now. Notice that showing the following suffices:

$$\forall r_1, r_2 \in \mathbb{Q} \text{ with } r_1 < r_2, \quad \exists x \in \mathbb{R} \setminus \mathbb{Q} \quad \text{such that} \quad r_1 < x < r_2.$$

By the Archimedean property, there exists $n \in \mathbb{N}_{>0}$ such that

$$\frac{\sqrt{2}}{n} < \frac{r_2 - r_1}{2}.$$

Then

$$\underbrace{r_1}_{\mathbb{Q}} < \underbrace{r_1 + \frac{\sqrt{2}}{n}}_{\mathbb{R}\setminus\mathbb{Q}} < \underbrace{r_2}_{\mathbb{Q}}.$$

For arbitrary real endpoints, the Archimedean property gives rational numbers r_1, r_2 between two given reals x, y.

□

Example

Let

$$A = \{r^n\}_{n=1}^{\infty}, \qquad r \in (0, 1).$$

Then $\inf A = 0$. As a consequence,

$$\forall \varepsilon > 0, \quad \exists N \text{ such that } \forall n > N, \quad r^n \leq \varepsilon.$$

Proof

Notice

$$\inf(rA) = r \inf(A).$$

Moreover,

$$rA \subset A \implies \inf(rA) \geq \inf A.$$

Since $r \in (0, 1)$, it follows that $\inf(A) = 0$. If there were no such N, then ε would be a positive lower bound. □

Example

Let

$$A := \left\{ \left(1 + \frac{1}{n}\right)^k \right\}_{n \geq 1}^{\infty}, \qquad k \in \mathbb{N}.$$

Then $\inf A = 1$.

Proof

By induction,

$$(1 + \alpha)^k \leq 1 + 2k\alpha, \qquad \forall \alpha \in \left[0, \frac{1}{2k}\right], \quad k \in \mathbb{N}.$$

Thus

$$1 \leq \left(1 + \frac{1}{n}\right)^k \leq 1 + \frac{2k}{n}.$$

By the Archimedean property, $2k/n$ can be made small. Therefore $\inf A = 1$ by the equivalent condition. □

Key Proof of the Least-Upper-Bound Property

Choose $a_0 \in A$, and let M be an upper bound of A. For each $n \in \mathbb{N}_{>0}$, the Archimedean property gives integers $K, L \in \mathbb{Z}$ such that K/n is an upper bound of A and L/n is not:

$$\frac{K}{n} \geq M, \qquad \frac{L}{n} < a_0.$$

Then, exhausting all the values from L to K, there exists $m_n \in \mathbb{Z}$ such that

$$\frac{m_n}{n} \text{ is an upper bound}, \qquad \frac{m_n - 1}{n} \text{ is not.}$$

This is true for every $n \in \mathbb{N}_{>0}$. We can verify that $\{m_n/n\}_{n=1}^{\infty}$, constructed like this, is Cauchy. Then we can verify, using the order and algebraic operations defined on $\mathbb{R}$, that

$$\left\{ \frac{m_n}{n} \right\}_{n=1}^{\infty}$$

is the least upper bound. □

Theorem

There exists $c \in \mathbb{R}$ such that $c^2 = 2$.

Proof

By the least-upper-bound property of $\mathbb{R}$, the supremum

$$c := \sup\{x \in \mathbb{R} \mid x > 0 \text{ and } x^2 \leq 2\}$$

exists in $\mathbb{R}$. Suppose $c^2 > 2$. Choose

$$\varepsilon := \frac{c^2 - 2}{2c}.$$

We can verify that $c - \varepsilon$ is a smaller upper bound,
so c is not the supremum, a contradiction.
Suppose $c^2 < 2$. Choose δ such that

$$0 < \delta < \min\left(1, \frac{2 - c^2}{2c + 1}\right).$$

Since $\delta < 1$,

$$(c + \delta)^2 = c^2 + 2c\delta + \delta^2 < c^2 + (2c + 1)\delta < 2.$$

Thus $c + \delta$ belongs to the set defining c, so c is not an upper bound, a contradiction. Therefore

$$c^2 = 2.$$

□

Another proof of the irrationality of $\sqrt{2}$

Suppose $\sqrt{2} \in \mathbb{Q}$. There exist $p, q \in \mathbb{N}_{>0}$ such that $\sqrt{2} = p/q$. Let

$$S = \{m \in \mathbb{N}_{>0} \mid m\sqrt{2} \in \mathbb{N}\}.$$

Then $S \neq \varnothing$ because $q \in S$. By the well-ordering principle on $\mathbb{N}$, there exists a least element $k_0 \in S$. Now let

$$k_1 := k_0\sqrt{2} - k_0.$$

Notice that $1 < \sqrt{2} < 2$. Thus

$$k_0 - k_0 < k_1 < 2k_0 - k_0.$$

Thus $k_1 \in \mathbb{N}_{>0}$ and $k_1 < k_0$. Also,

$$k_1\sqrt{2} = (k_0\sqrt{2} - k_0)\sqrt{2} = 2k_0 - \sqrt{2}k_0 \in \mathbb{Z},$$

so $k_1 \in S$. But $k_1 < k_0$, a contradiction. Thus $\sqrt{2} \notin \mathbb{Q}$. □

Theorem

For every $k \in \mathbb{N}$ and $x \in (0,1)$, let

$$a_n := n^k x^n.$$

For every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that, for every $n \geq N$,

$$a_n \leq \varepsilon.$$

Proof

$$\frac{a_{n+1}}{a_n} = \left(1 + \frac{1}{n}\right)^k x, \qquad \forall n \in \mathbb{N}^+.$$

We know

$$\inf\left\{\left(1 + \frac{1}{n}\right)^k : n \geq 1\right\} = 1,$$

and the sequence is decreasing. Put

$$b_n := \left(1 + \frac{1}{n}\right)^k$$

and choose

$$\delta \in \left(0, \frac{1}{2}\left(\frac{1}{x} - 1\right)\right).$$

Since $\inf\{b_n\} = 1$, there exists $N \in \mathbb{N}$ such that $b_N < 1 + \delta$. Since $\{b_n\}$ is decreasing, $b_n \leq b_N$ for every $n \geq N$; thus

$$b_n < 1 + \delta, \qquad \forall n \geq N.$$

If

$$r := (1 + \delta)x,$$

then $r < 1$ by the choice of δ. Moreover,

$$0 < \frac{a_{n+1}}{a_n} \leq r < 1, \qquad \forall n \geq N.$$

Thus

$$0 < a_n \leq r a_{n-1} \leq \cdots \leq r^{n-N} a_N, \qquad \forall n \geq N.$$

Since $r^n \to 0$, there exists $N_2 \in \mathbb{N}$ such that

$$r^n < \frac{\varepsilon r^N}{a_N}, \qquad \forall n \geq N_2.$$

Choose $N_1 := \max(N, N_2)$. Then, for every $n \geq N_1$,

$$a_n \leq r^{n-N} a_N < \frac{\varepsilon r^N}{a_N} a_N r^{-N} = \varepsilon.$$

□

2.3 Powers and Roots

Definition

Let $x \in \mathbb{R}_{\geq 0}$ and $n \in \mathbb{N}_{>0}$. The nth root of x is

$$x^{1/n} := \sup\{z \in \mathbb{R} \mid z \geq 0,\ z^n \leq x\}.$$

By convention, for every $x \in \mathbb{R}$,

$$x^0 := 1 \qquad \text{(including 0)}.$$

Lemma

For every $x, y \geq 0$ and $m, n \in \mathbb{N}_{>0}$:

1. $x^{1/n} \in \mathbb{R}^{\geq 0}$ and $(x^{1/n})^n = x$.
2. $x < y \iff x^{1/n} < y^{1/n}$.
3. $(xy)^{1/n} = x^{1/n} y^{1/n}$.
4. $(x^{1/n})^{1/m} = x^{1/(nm)}$.

The proof is trivial. □

Lemma

Let $x \in \mathbb{R}^{>0}$, $m, \widetilde{m} \in \mathbb{Z}$, and $n, \widetilde{n} \in \mathbb{N} \setminus \{0\}$. Then

$$\frac{m}{n} = \frac{\widetilde{m}}{\widetilde{n}} \implies (x^{1/n})^m = (x^{1/\widetilde{n}})^{\widetilde{m}}.$$

Proof

Suppose $m > 0$; then $\widetilde{m} > 0$. We have

$$\begin{aligned} \frac{m}{n} = \frac{\widetilde{m}}{\widetilde{n}} &\iff n\widetilde{m} = m\widetilde{n} \\ &\iff \frac{1}{n\widetilde{m}} = \frac{1}{m\widetilde{n}} \\ &\implies x^{1/(n\widetilde{m})} = x^{1/(m\widetilde{n})} \\ &\implies \left(x^{1/(n\widetilde{m})}\right)^{m\widetilde{m}} = \left(x^{1/(m\widetilde{n})}\right)^{m\widetilde{m}}. \end{aligned}$$

Then apply the preceding lemma. If $m < 0$, apply what we proved to $-m$ and $-\widetilde{m}$, then take the reciprocal since $x > 0$. If $m = 0$, then $1 = 1$, the trivial case. □

Definition

An open interval and a closed interval are, respectively,

$$(a, b) := \{x \in \mathbb{R} \mid a < x < b\}, \qquad [a, b] := \{x \in \mathbb{R} \mid a \leq x \leq b\}.$$

2.4 The Nested-Interval Property

The following completeness properties are equivalent:

$$\text{Nested Interval Property of } \mathbb{R} \iff \underbrace{\text{least-upper-bound property of } \mathbb{R}}_{\substack{\text{proven from the} \\ \text{construction of } \mathbb{R}}}$$

$$\iff \underbrace{\text{Bolzano–Weierstrass theorem}}_{\substack{\text{every bounded sequence has} \\ \text{a convergent subsequence}}} \iff \underbrace{\text{Cauchy complete}}_{\substack{\text{our construction} \\ \text{of } \mathbb{R}}}.$$

Key Theorem

Let

$$I_n = [a_n, b_n], \qquad a_n < b_n, \qquad n \in \mathbb{N},$$

such that

$$\cdots \subset I_{n+1} \subset I_n \subset \cdots \subset I_2 \subset I_1.$$

Then

$$\bigcap_{n=1}^{\infty} I_n \neq \varnothing.$$

Remark

If I_n is not closed, this may not be true:

$$\bigcap_{n=1}^{\infty} \left(1 - \frac{1}{n}, 1\right) = \varnothing.$$

Proof

Let

$$A := \{a_n\}_{n=1}^{\infty}, \qquad B := \{b_n\}_{n=1}^{\infty}.$$

Notice that A is bounded above by any element of B, and B is bounded below by any element of A. Thus it becomes trivial to show that

$$\sup A \leq \inf B,$$

and

$$\varnothing \neq [\sup A, \inf B] \subset \bigcap_{n=1}^{\infty} [a_n, b_n].$$

□

2.5 Uncountability and Interval Bijections

Theorem

$\mathbb{R}$ is uncountable.

Proof (an alternate proof from Cantor's diagonal argument)

Suppose there exists a bijection $f : \mathbb{N} \to \mathbb{R}$. Then

$$\mathbb{R} = \{x_n\}_{n=1}^{\infty}.$$

Set

$$I_1 := [x_1 + 1, x_1 + 2]$$

and, inductively, divide I_n into three closed subintervals and choose $I_{n+1} \subset I_n$ so that $x_{n+1} \notin I_{n+1}$. Such a choice is always possible because at least one of the three subintervals omits x_{n+1}. Thus $x_n \notin I_n$ for every n.
By the Nested Interval Property, $\bigcap_{n=1}^{\infty} I_n$ is nonempty. If y belonged to this intersection, then $y = x_j$ for some j, because the sequence enumerates $\mathbb{R}$; but $x_j \notin I_j$, a contradiction. Hence $\mathbb{R}$ is uncountable. □

Corollary

The irrationals are uncountable.

Proof

Suppose $\mathbb{R} \setminus \mathbb{Q}$ is countable. Then

$$\mathbb{R} = (\mathbb{R} \setminus \mathbb{Q}) \cup \mathbb{Q},$$

which is a union of two countable sets. Hence $\mathbb{R}$ is countable, a contradiction. □

Corollary

(a, b) is uncountable.

Proof

Let

$$r = \frac{b-a}{2}, \qquad c = \frac{b+a}{2}.$$

Then

$$f(x) = \frac{x-c}{r^2 - (x-c)^2}$$

is a bijection from (a, b) to $\mathbb{R}$. The denominator is the product of the distances to the endpoints, so the formula may be viewed as

$$\frac{\text{distance to midpoint}}{\text{product of distances to endpoints}},$$

with values tending to $-\infty$ near the left endpoint and $+\infty$ near the right endpoint. □

Proposition

There exists a bijection

$$f : [a, b) \longrightarrow (a, b).$$

Proof

$$(a, b) = \left[(\mathbb{R} \setminus \mathbb{Q}) \cap (a, b)\right] \cup \underbrace{\left(\mathbb{Q} \cap (a, b)\right)}_{\text{countable}}.$$

Enumerate

$$\mathbb{Q} \cap (a, b) = \{r_1, r_2, \ldots\}.$$

Define $f : [a, b) \to (a, b)$ as follows. If $x \in \{a\} \cup (\mathbb{Q} \cap (a, b))$, set

$$x = a \implies f(x) = r_1, \qquad x \neq a \implies f(r_i) = r_{i+1}.$$

If $x \in (\mathbb{R} \setminus \mathbb{Q}) \cap (a, b)$, set $f(x) = x$. Then f is bijective. □

2.6 Cantor–Schröder–Bernstein

Theorem

If $|X| \leq |Y|$ and $|Y| \leq |X|$, that is, there exist injections $X \to Y$ and $Y \to X$, then

$$|X| = |Y|.$$

Lemma

Let

$$\varphi : \mathcal{P}(X) \longrightarrow \mathcal{P}(X)$$

be a function with monotonicity

$$A_1 \subset A_2 \implies \varphi(A_1) \subset \varphi(A_2).$$

Then there exists $A_0 \in \mathcal{P}(X)$ such that $\varphi(A_0) = A_0$.

Proof

Consider

$$D := \{A \in \mathcal{P}(X) \mid A \subset \varphi(A)\}.$$

Define

$$A_0 := \bigcup_{A \in D} A.$$

Then

$$A_0 = \bigcup_{A \in D} A \underset{\text{definition of } D}{\subset} \bigcup_{A \in D} \varphi(A) \underset{\text{monotonicity}}{\subset} \varphi(A_0).$$

Moreover,

$$\varphi(A_0) \subset \varphi(\varphi(A_0))$$

by monotonicity, so $\varphi(A_0) \in D$. Since $A_0 = \bigcup_{A \in D} A$, it follows that

$$\varphi(A_0) \subset A_0,$$

and hence $\varphi(A_0) = A_0$. □

Lemma

If there exist injections $f : X \to Y$ and $g : Y \to X$, then there exists $A_0 \subset X$ such that

$$g^{-1} : X \setminus A_0 \longrightarrow Y \setminus f(A_0)$$

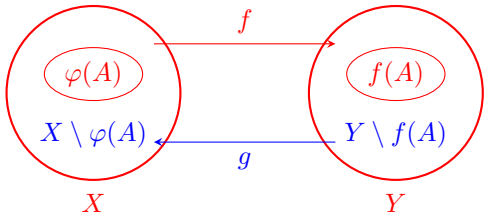
is bijective.

Proof

Define

$$\varphi : \mathcal{P}(X) \longrightarrow \mathcal{P}(X), \qquad \varphi(A) := X \setminus g(Y \setminus f(A)).$$

The maps and complementary subsets are related as follows:



We claim that φ is monotone. Let $A_1 \subset A_2 \subset X$. Then

$$f(A_1) \subset f(A_2) \implies Y \setminus f(A_1) \supset Y \setminus f(A_2) \implies g(Y \setminus f(A_1)) \supset g(Y \setminus f(A_2)).$$

Therefore

$$\varphi(A_1) = X \setminus g(Y \setminus f(A_1)) \subset X \setminus g(Y \setminus f(A_2)) = \varphi(A_2).$$

Thus, by the fixed-point lemma, there exists $A_0 \in \mathcal{P}(X)$ such that $\varphi(A_0) = A_0$, that is,

$$X \setminus g(Y \setminus f(A_0)) = A_0.$$

Thus

$$X \setminus A_0 = g(Y \setminus f(A_0)).$$

The map g is surjective from $Y \setminus f(A_0)$ to $X \setminus A_0$, so

$$g^{-1} : X \setminus A_0 \longrightarrow Y \setminus f(A_0)$$

is bijective. □

Proof of Cantor–Schröder–Bernstein

Notice that

$$f : A_0 \longrightarrow f(A_0)$$

is bijective, and

$$g^{-1} : X \setminus A_0 \longrightarrow Y \setminus f(A_0)$$

is bijective. Thus we have a bijection from X to Y. □

3 Sequences and Series

3.1 Limits of Sequences

Examples

$$\forall s > 0, \qquad \lim_{n\to\infty} \frac{1}{n^s} = 0,$$
$$\lim_{n\to\infty} \frac{1}{2^{sn}} = 0,$$
$$\lim_{n\to\infty} \frac{n!}{n^n} = 0,$$
$$\lim_{n\to\infty} \frac{\sum_{k=1}^{n} k}{n^2} = \frac{1}{2},$$
$$\lim_{n\to\infty} \frac{(-1)^n n}{n+1} \quad \text{does not exist.}$$

Theorem

The limit of a convergent sequence is unique.

Theorem

If $\{x_n\}_n$ converges, then it is bounded.

Theorem

Let $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}_{n=1}^{\infty}$ satisfy

$$\lim_{n\to\infty} x_n = x, \qquad \lim_{n\to\infty} y_n = y.$$

Then

$$\forall c \in \mathbb{R}, \qquad \lim_{n\to\infty} cx_n = cx,$$
$$\lim_{n\to\infty} (x_n \pm y_n) = \lim_{n\to\infty} x_n \pm \lim_{n\to\infty} y_n,$$
$$\lim_{n\to\infty} (x_n y_n) = \left(\lim_{n\to\infty} x_n\right)\left(\lim_{n\to\infty} y_n\right),$$
$$y \neq 0,\ y_n \neq 0 \implies \lim_{n\to\infty} \frac{x_n}{y_n} = \frac{x}{y}.$$

Proposition

If $x_n \geq 0$ for every n and $\lim_{n\to\infty} x_n = x$, then $x \geq 0$.
If $x_n \geq y_n$ for every n, $\lim_{n\to\infty} x_n = x$, and $\lim_{n\to\infty} y_n = y$, then $x \geq y$.

Key Proposition

For every $s > 0$,

$$\lim_{n\to\infty} s^{1/n} = 1.$$

Proof

The case $s = 1$ is trivial. If $s > 1$, then

$$\left[1 + (s^{1/n} - 1)\right]^n \geq 1 + n(s^{1/n} - 1).$$

Therefore

$$0 \leq s^{1/n} - 1 \leq \frac{s-1}{n},$$

and the result follows from the squeeze theorem. If $s < 1$, then

$$\lim_{n\to\infty} s^{1/n} = \lim_{n\to\infty} \frac{1}{(1/s)^{1/n}} = \frac{1}{1} = 1.$$

□

Key Proposition

$$\lim_{n\to\infty} n^{1/n} = 1.$$

Proof

Put $x_n = n^{1/n} - 1$. By the binomial expansion,

$$n = (1 + n^{1/n} - 1)^n = \sum \cdots \geq \frac{n(n-1)}{2} x_n^2 = \binom{n}{2} x_n^2.$$

Hence

$$x_n \leq \sqrt{\frac{2}{n-1}}, \qquad 0 \leq x_n \leq \sqrt{\frac{2}{n-1}},$$

and therefore, by the squeeze theorem,

$$n^{1/n} - 1 \xrightarrow[n\to\infty]{} 0.$$

□

Theorem (Squeeze Theorem)

Let $\{x_n\}$, $\{y_n\}$, and $\{z_n\}$ be sequences such that, for all sufficiently large n,

$$x_n \leq y_n \leq z_n.$$

If $x_n \to a$ and $z_n \to a$, then

$$y_n \to a.$$

Proof

If the common limit is a, then, eventually,

$$a - \varepsilon < x_n \leq y_n \leq z_n < a + \varepsilon,$$

so $\lim_{n\to\infty} y_n = a$. □

Definition

A sequence $\{x_n\}_{n=1}^{\infty}$ is

monotone increasing	if $x_n \leq x_{n+1}$,
monotone decreasing	if $x_n \geq x_{n+1}$,
bounded below	if $\exists M \in \mathbb{R}$ such that $x_n \geq M$,
bounded above	if $\exists M \in \mathbb{R}$ such that $x_n \leq M$,
bounded	if $\exists M \in \mathbb{R}$ such that $\lvert x_n\rvert \leq M$.

Theorem (Monotone Convergence Theorem)

If $\{x_n\}_{n=1}^{\infty}$ is monotone increasing and bounded above, then its limit exists (the sequence converges). If it is monotone decreasing and bounded below, then its limit exists (the sequence converges).

Proof

The supremum, respectively the infimum, is the limit; existence follows from the least-upper-bound property. □

Example

Let

$$x_1 = \sqrt{2}, \qquad x_{n+1} = \sqrt{2 + x_n}.$$

Find $\lim_{n\to\infty} x_n$. We have $x_{n+1} \geq x_n$ by mathematical induction and $x_n \leq 2$ by induction.

Definition

Let $\{x_n\}_{n=1}^{\infty}$ be bounded.

$$\limsup_{n\to\infty}(x_n) := \lim_{n\to\infty} \overline{x}_n = \lim_{n\to\infty} \sup_{k\geq n}\{x_k\} = \inf_{n\in\mathbb{N}} \sup_{k\geq n}\{x_k\},$$

$$\liminf_{n\to\infty}(x_n) := \lim_{n\to\infty} \underline{x}_n = \lim_{n\to\infty} \inf_{k\geq n}\{x_k\} = \sup_{n\in\mathbb{N}} \inf_{k\geq n}\{x_k\}.$$

Theorem

If $\{x_k\}_{k=1}^{\infty}$ is bounded, then $\limsup x_n$ and $\liminf x_n$ exist.

Proof

The sequence

$$\left\{\inf_{k\geq n} x_k\right\}_{n=1}^{\infty}$$

is monotone increasing and bounded above by $\sup\{x_n\}$, so the monotone convergence theorem applies. The same argument applies to $\left\{\sup_{k\geq n} x_k\right\}_n$. □

Theorem

Let $\{x_n\}$ be bounded. Then

$$\{x_n\} \text{ is convergent} \iff \limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n.$$

If so, then

$$\lim x_n = \limsup x_n = \liminf x_n.$$

Proof

For ($\Leftarrow$), suppose

$$L = \liminf x_n = \limsup x_n.$$

For every $n \in \mathbb{N}$,

$$\inf\{x_k : k \geq n\} \leq x_n \leq \sup\{x_k : k \geq n\},$$

and both outer terms tend to L. By the squeeze theorem, $x_n \to L$.
For ($\Rightarrow$), Bolzano–Weierstrass gives a shorter proof. For a direct argument, suppose $\lim_{n\to\infty} x_n = x$. We show

$$\lim_{n\to\infty} \sup_{k\geq n} x_k = \lim_{n\to\infty} \inf_{k\geq n} x_k = x.$$

There exists $N \in \mathbb{N}$ such that, for every $n \geq N$, $|x_n - x| < \varepsilon$. Hence

$$x \leq \sup_{k\geq n} x_k < x + \varepsilon, \qquad x - \varepsilon < \inf_{k\geq n} x_k \leq x,$$

and consequently both tail extrema tend to x. □

Proposition (Liminf and limsup inequalities)

Let $\{x_n\}_{n=1}^{\infty}$ and $\{y_n\}$ be bounded. Then, for $\alpha \in \mathbb{R}_{\geq 0}$,

$$\begin{aligned}
\liminf(\alpha x_n) &= \alpha \liminf x_n, \\
\limsup(\alpha x_n) &= \alpha \limsup x_n, \\
\liminf x_n + \liminf y_n &\leq \liminf(x_n + y_n), \\
\limsup x_n + \limsup y_n &\geq \limsup(x_n + y_n).
\end{aligned}$$

Definition (Cauchy sequence in $\mathbb{R}$)

A sequence $\{x_n\}_{n=1}^{\infty} \subset \mathbb{R}$ is Cauchy if

$$\forall \varepsilon > 0 \; \exists N > 0 \text{ (or } N \in \mathbb{N}) \quad \text{such that} \quad \forall k, \ell \geq N, \quad |x_k - x_\ell| < \varepsilon.$$

Proposition

A sequence $\{x_n\}_{n=1}^{\infty} \subset \mathbb{R}$ is Cauchy if and only if it is convergent.

Proof

One proof uses Bolzano–Weierstrass: a Cauchy sequence is bounded and therefore has a convergent subsequence; the Cauchy property then forces the entire sequence to converge. Alternatively, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that, for all $n, \ell \geq N$,

$$|x_n - x_\ell| < \frac{\varepsilon}{2}.$$

Thus

$$0 \leq \sup_{n \geq N} x_n - \inf_{n \geq N} x_n \leq \frac{\varepsilon}{2} < \varepsilon,$$

so $\limsup x_n = \liminf x_n$, and the sequence converges by the liminf–limsup criterion. □

Definition

A sequence $\{x_n\} \subset \mathbb{R}$ is *contractive* if there is a contractive constant $K \in (0, 1)$ such that

$$|x_{n+2} - x_{n+1}| \leq K|x_{n+1} - x_n|, \qquad \forall n \in \mathbb{N}.$$

Proposition

If $\{x_n\}$ is contractive, then it converges in $\mathbb{R}$.

Proof

Show that $\{x_n\}$ is Cauchy. □

Counterexample: convergence does not imply contractiveness

The sequence $\{1/n\}_{n=1}^{\infty}$ converges, but there is no $K \in (0, 1)$ such that

$$|x_{n+2} - x_{n+1}| \leq K|x_{n+1} - x_n|.$$

Example

Let

$$x_1 = 1, \qquad x_2 = 2, \qquad x_{n+1} = \frac{1}{2}(x_n + x_{n-1}).$$

Then

$$x_{n+1} - x_n =: y_n = \frac{1}{2}(x_n + x_{n-1}) - x_n = \frac{1}{2}(x_{n-1} - x_n) = -\frac{1}{2}y_{n-1}.$$

Since $y_1 = x_2 - x_1 = 1$,

$$y_n = \left(-\frac{1}{2}\right)^{n-1}.$$

Also,

$$\begin{aligned} x_n &= (x_n - x_{n-1}) + (x_{n-1} - x_{n-2}) + \cdots + (x_2 - x_1) + x_1 \\ &= \sum_{i=1}^{n-1} y_i + x_1 = \sum_{i=1}^{n-1} \left(-\frac{1}{2}\right)^{i-1} + 1. \end{aligned}$$

Therefore

$$\lim_{n\to\infty} x_n = \frac{1}{1-(-1/2)} + 1 = \frac{2}{3} + 1 = \frac{5}{3}.$$

3.2 Euler's Number e

Proposition

Let

$$x_n = \left(1 + \frac{1}{n}\right)^n, \qquad y_n = \left(1 + \frac{1}{n}\right)^{n+1}.$$

Then $\{x_n\}_n$ is strictly increasing and $\{y_n\}_n$ is strictly decreasing.

Proof

Using $(1+x)^n \geq 1 + nx$ for $x > -1$,

$$\begin{aligned} \frac{x_{n+1}}{x_n} &= \frac{n+2}{n+1}\left(1 - \frac{1}{(n+1)^2}\right)^n \\ &\geq \frac{n+2}{n+1}\left(1 - \frac{n}{(n+1)^2}\right) = \frac{n^3+3n^2+3n+2}{n^3+3n^2+3n+1} > 1. \end{aligned}$$

Furthermore,

$$\begin{aligned} \frac{y_n}{y_{n+1}} &= \frac{n+1}{n+2}\left(1 + \frac{1}{n^2+2n}\right)^{n+1} \\ &\geq \frac{n+1}{n+2}\left(1 + \frac{n+1}{n^2+2n}\right) = \frac{n^3+4n^2+4n+1}{n^3+4n^2+4n} > 1. \end{aligned}$$

Thus $y_n > y_{n+1}$. □

Proposition

For all $n, k \in \mathbb{N}$, $x_n < y_k$.

Proof

By definition, $x_n < y_n$. If $n > k$, then

$$x_n < y_n < y_k.$$

If $n \leq k$, then

$$x_n \leq x_k < y_k.$$

□

Corollary

The sequence $\{x_n\}$ is bounded above, $\{y_k\}$ is bounded below, and

$$\lim_n x_n = \lim_n y_n.$$

Proof

By the monotone convergence theorem, the limits exist, and

$$\lim_{n\to\infty} \frac{y_n}{x_n} = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right) = 1 = \frac{\lim y_n}{\lim x_n}.$$

□

3.3 Bolzano–Weierstrass

Theorem (Bolzano–Weierstrass)

Every bounded sequence has a convergent subsequence.

Proof

We use the nested-interval property of $\mathbb{R}$. An alternative proof uses the least-upper-bound property.
Since $\{x_n\}$ is bounded, there exists $M > 0$ such that

$$\{x_n\} \subset [-M, M].$$

Set $I_0 = [-M, M]$. Bisect I_0 and choose a closed half containing infinitely many terms of the sequence. Repeat this construction to obtain

$$I_k = [a_k, b_k], \qquad \cdots \subset I_{k+1} \subset I_k \subset \cdots \subset I_0,$$

where each I_k contains infinitely many terms of $\{x_n\}$ and

$$b_k - a_k = \frac{2M}{2^k}.$$

Choose the indices recursively so that $n_1 < n_2 < \cdots$ and $x_{n_k} \in I_k$. This is possible because every I_k contains infinitely many terms. By the nested-interval property, there exists

$$c \in \bigcap_{k=0}^{\infty} I_k.$$

Hence

$$|x_{n_k} - c| \le b_k - a_k = \frac{2M}{2^k} \longrightarrow 0,$$

so $x_{n_k} \to c$. □

Definition

Let $A \subset \mathbb{R}$. A point $x \in \mathbb{R}$ is a *limit point* (cluster point) of A if

$$\forall \varepsilon > 0 \; \exists y \in A \quad \text{such that} \quad 0 < |x - y| < \varepsilon.$$

Note

A limit point of A need not belong to A.

Definition

A set $A \subset \mathbb{R}$ is *closed* if every limit point of A belongs to A.

Definition

The closure of A is

$$\overline{A} := A \cup \{\text{limit points of } A\}.$$

Lemma

$\overline{A}$ is closed.

Proof

Let $a \in \mathbb{R}$ be a limit point of $\overline{A}$. If $a \in A$, we are done. Suppose $a \notin A$. For every $\varepsilon > 0$, there exists $b \in \overline{A}$ such that

$$0 < |a - b| < \frac{\varepsilon}{2}.$$

If $b \in A$, then a is a limit point of A, so $a \in \overline{A}$. If $b \notin A$, then b is a limit point of A; hence there exists $c \in A$ such that

$$0 < |b - c| < \frac{\varepsilon}{2}.$$

Because $a \notin A$ and $c \in A$, we have $c \neq a$, and

$$0 < |a - c| \leq |a - b| + |b - c| < \varepsilon.$$

Thus a is a limit point of A, and therefore $a \in \overline{A}$. □

Definition

A set $A \subset \mathbb{R}$ is (sequentially) compact if every sequence $\{x_n\}_n \subset A$ has a convergent subsequence that converges in A; that is,

$$\lim_{k \to \infty} x_{n_k} = x \in A.$$

Theorem (Heine–Borel)

A set $A \subset \mathbb{R}$ is compact if and only if it is closed and bounded.

Proof

For ($\Leftarrow$), apply Bolzano–Weierstrass.
For ($\Rightarrow$), suppose A is unbounded. Choose $x_n \in A$ so that $|x_n| > n$ for every n. Every subsequence $\{x_{n_k}\}$ is unbounded because $|x_{n_k}| > n_k \to \infty$, so no subsequence can converge. This contradicts sequential compactness. Hence A is bounded.
To see that A is closed, let $x \in \mathbb{R}$ be a limit point of A. Then there is $\{x_n\} \subset A$ such that

$$\lim x_n = x \in \mathbb{R}.$$

By compactness, there is a subsequence $\{x_{n_k}\}$ such that $x_{n_k} \to y \in A$. A subsequence has the same limit as the convergent sequence, so $x = y \in A$. Hence A is closed. □

Theorem (Heine–Borel, open-cover form)

For $A \subset \mathbb{R}^n$, A is compact in the open-cover sense if and only if A is closed and bounded.

Proof

To show that the Heine–Borel property implies boundedness of A, notice that, for $\varepsilon > 0$,

$$\bigcup_{x \in A} U(x, \varepsilon)$$

is an open cover of A, and therefore contains a finite subcover

$$A \subset \bigcup_{i=1}^{N} U(x_i, \varepsilon).$$

Define

$$M := \sum_{i=2}^{N} d(x_1, x_i).$$

By the triangle inequality, it is immediate that $A \subset U(x_1, M + \varepsilon)$.
To show that A is closed, we show that A^c is open. Let $x_0 \in A^c$, and define

$$C_N := \left\{ x \in \mathbb{R}^n : d(x, x_0) \leq \frac{1}{N} \right\}, \qquad U_N := (C_N)^c.$$

Then

$$\bigcap_{N=1}^{\infty} C_N = \{x_0\}, \qquad \bigcup_{N=1}^{\infty} U_N = \mathbb{R}^n \setminus \{x_0\}$$

by De Morgan's law. Since $x_0 \in A^c$, hence $x_0 \notin A$, we have $A \subset \bigcup_{N=1}^{\infty} U_N$. This is an open cover of A, so the Heine–Borel property yields a finite subcover $\{U_{N_1}, \ldots, U_{N_k}\}$.
Without loss of generality, take $N_1 < N_2 < \cdots < N_k$. By the definition of the U_N's,

$$U_{N_1} \subset U_{N_2} \subset \cdots \subset U_{N_k}.$$

Thus

$$A \subset \bigcup_{i=1}^{k} U_{N_i} = U_{N_k} \quad \Longrightarrow \quad (U_{N_k})^c = C_{N_k} \subset A^c.$$

Since $U(x_0, 1/N_k) \subset C_{N_k} \subset A^c$, the set A^c is open, and hence A is closed.
For the converse, first consider the n-box centered at the origin,

$$Q = [-a, a] \times \cdots \times [-a, a] \subset \mathbb{R}^n.$$

We claim that Q has the Heine–Borel property. Suppose, for contradiction, that Q does not have the Heine–Borel property, and let W be an open cover of Q having no finite subcover. Bisect each side of the box and connect the midpoints, obtaining 2^n sub-boxes. At least one sub-box cannot be covered by finitely many open sets in W; call it Q_1. Bisect Q_1 in the same way, and continue. This yields nested boxes

$$Q \supset Q_1 \supset Q_2 \supset \cdots,$$

none of which can be covered by finitely many open sets from W.
Choose $c_j \in Q_j$ for every $j \geq 1$. The box Q_j has side length $2a/2^j$ and Euclidean diameter $2a\sqrt{n}/2^j$. If $k \geq j$, then $c_k, c_j \in Q_j$, so

$$\|c_k - c_j\|_2 \leq \operatorname{diam}(Q_j) = \frac{2a\sqrt{n}}{2^j} \longrightarrow 0.$$

Thus $\{c_j\}$ is Cauchy. Since $\mathbb{R}^n$ is complete, there is $c \in \mathbb{R}^n$ such that $c_j \to c$. Moreover, $c \in Q_j$ for every $j \geq 1$.
There is some $K \in W$ with $c \in K$. Since K is open, there is $\varepsilon > 0$ such that $U(c, \varepsilon) \subset K$. For N large enough,

$$Q_N \subset U(c, \varepsilon) \subset K.$$

But $K \in W$, while Q_N was constructed so that it cannot be covered by finitely many members of W, a contradiction. Therefore Q has the Heine–Borel property.
Finally, let $A \subset \mathbb{R}^n$ be closed and bounded. Since A is bounded, there is a box $Q = [-a, a]^n$, centered at the origin and with side length $2a$, such that $A \subset Q$. Since A is closed, $A^c = \mathbb{R}^n \setminus A$ is open. If $\mathcal{A}$ is an open cover of A, then

$$\mathcal{A} \cup \{\mathbb{R}^n \setminus A\}$$

is an open cover of Q. By the preceding part it has a finite subcover $\mathcal{K}$. Removing $\mathbb{R}^n \setminus A$, if present, leaves a finite subcover of $\mathcal{A}$ that covers A. Thus A is compact in the open-cover sense. □

3.4 Proper Divergence

Definition

We say

$$x_n \longrightarrow \infty \quad \text{if} \quad \forall M > 0 \; \exists N \in \mathbb{N} \text{ such that } \forall n \geq N, \; x_n > M,$$

and

$$x_n \longrightarrow -\infty \quad \text{if} \quad \forall M > 0 \; \exists N \in \mathbb{N} \text{ such that } \forall n \geq N, \; x_n < -M.$$

We say that $\lim_{n\to\infty} x_n$ exists in

$$\mathbb{R} \cup \{\pm\infty\},$$

the extended reals.

Proposition

lim sup and lim inf always exist in

$$\overline{\mathbb{R}} = \mathbb{R} \cup \{\pm\infty\}.$$

If $\lim x_n$ exists in $\overline{\mathbb{R}}$, then

$$\liminf x_n = \limsup x_n = \lim x_n$$

is still true.

3.5 Series

Let $\{x_n\} \subset \mathbb{R}$.

Definition

We call

$$\sum_{k=1}^{\infty} x_k$$

a *formal series*, for which we do not care about convergence at all. Let

$$S_n := \sum_{k=1}^{n} x_k$$

be the nth partial sum of this formal series.
If $\{S_n\}_n$ converges in $\mathbb{R}$, then we say that the formal series

$$\sum_{k=1}^{\infty} x_k$$

converges in $\mathbb{R}$, and it converges to $\lim_{n\to\infty} S_n$.
We say that a series is *absolutely convergent* if

$$\sum_{k} |x_k|$$

converges in $\mathbb{R}$.

Lemma

$$\sum_{k=1}^{\infty} x_k \text{ converges absolutely} \implies \sum_{k=1}^{\infty} x_k \text{ converges.}$$

Proof

Use the triangle inequality. □

Lemma

$$\sum_k x_k \text{ converges} \iff \begin{array}{c}\forall \varepsilon > 0\ \exists N \in \mathbb{N} \text{ such that} \\ \forall m > n \geq N, \quad \left| \sum_{k=n+1}^{m} x_k \right| < \varepsilon\end{array}.$$

Proof

Note that

$$\sum x_k \text{ converges} \iff S_n \text{ converges in } \mathbb{R} \iff S_n \text{ is Cauchy},$$

which is equivalent to the displayed tail condition. □

Corollary

If $\sum_k x_k$ converges, then

$$\lim_{n\to\infty} x_n = 0.$$

Proof

By the Cauchy criterion for series. □

Lemma

If $\{x_n\}_n \subset \mathbb{R}_{\geq 0}$, then

$$\sum_k x_k \text{ converges} \iff \{S_n\}_n \text{ is bounded above.}$$

Proof

By the monotone convergence theorem. □

Proposition (Comparison Test)

Suppose there is $N \in \mathbb{N}$ such that, for every $n \geq N$,

$$0 \leq a_n \leq b_n.$$

Then:

1. if $\sum b_n$ converges, then $\sum a_n$ converges;
2. if $\sum a_n$ diverges, then $\sum b_n$ diverges.

Proof

By contraposition. □

Example

For every $x \in (-1, 1)$,

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}.$$

Lemma (Cauchy Condensation)

If $\{x_n\}$ satisfies

$$x_1 \geq x_2 \geq \cdots \geq 0,$$

then

$$\sum_{n=1}^{\infty} x_n \text{ converges} \quad \Longleftrightarrow \quad \sum_{k=0}^{\infty} 2^k x_{2^k} \text{ converges.}$$

Proof

For ($\Leftarrow$), for every $K \geq 0$,

$$S_{2^{K+1}-1} = \sum_{j=0}^{K} \sum_{n=2^j}^{2^{j+1}-1} x_n \leq \sum_{j=0}^{K} 2^j x_{2^j}.$$

If the condensed series converges, the right-hand side is bounded independently of K. Since $\{S_n\}$ is increasing and the dyadic-endpoint subsequence $\{S_{2^{K+1}-1}\}$ is bounded, $\{S_n\}$ is bounded and therefore converges.
For ($\Rightarrow$), suppose $\sum_{n=1}^{\infty} x_n$ converges. For every $K \geq 1$,

$$\sum_{j=1}^{K} 2^{j-1} x_{2^j} \leq \sum_{j=1}^{K} \sum_{n=2^{j-1}+1}^{2^j} x_n \leq \sum_{n=1}^{2^K} x_n.$$

Consequently,

$$\sum_{j=0}^{K} 2^j x_{2^j} \leq x_1 + 2 \sum_{n=1}^{2^K} x_n.$$

The condensed partial sums are increasing and bounded, so the condensed series converges.

Remark

The preceding comparison of infinite series is valid because limits preserve weak inequalities. □

Theorem (p-series)

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \text{ converges} \quad \Longleftrightarrow \quad p > 1.$$

It diverges if $p \leq 1$.

Proof

If $p \leq 0$, the series diverges. Suppose $p > 0$.
Then

$$\sum_{k=0}^{\infty} 2^k \frac{1}{(2^k)^p} = \sum_{k=0}^{\infty} 2^{k(1-p)},$$

a geometric series. Thus the condensed series converges precisely when $p > 1$, and Cauchy condensation gives the result. □

Example

For every $p > 0$,

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^p}$$

converges. This is a special case of the Alternating Series Test.

Proof

Let

$$S_N := \sum_{n=1}^{N} \frac{(-1)^n}{n^p}.$$

Because n^{-p} decreases to 0,

$$S_{2m+2} - S_{2m} = -\frac{1}{(2m+1)^p} + \frac{1}{(2m+2)^p} \leq 0,$$

so $\{S_{2m}\}$ is decreasing. Likewise,

$$S_{2m+3} - S_{2m+1} = \frac{1}{(2m+2)^p} - \frac{1}{(2m+3)^p} \geq 0,$$

so $\{S_{2m+1}\}$ is increasing. Since

$$S_{2m+1} = S_{2m} - \frac{1}{(2m+1)^p} \leq S_{2m},$$

the odd partial sums are bounded above and the even partial sums are bounded below. Both subsequences therefore converge, and

$$S_{2m} - S_{2m+1} = \frac{1}{(2m+1)^p} \longrightarrow 0,$$

so they have the same limit. Thus the full sequence of partial sums converges. □

Theorem

Let

$$S_n := \sum_{k=0}^{n} \frac{1}{k!}.$$

Then $\{S_n\} \subset \mathbb{R}$ is Cauchy and

$$\boxed{\lim_{n\to\infty} S_n = e}.$$

As a consequence, for every $n \geq 1$,

$$0 < e - S_n < \frac{1}{n!\, n}.$$

Proof

Let

$$x_n = \left(1 + \frac{1}{n}\right)^n.$$

By the monotonicity result for the defining sequence of e, $\{x_n\}$ is increasing and

$$\lim_{n\to\infty} x_n = e.$$

By the binomial expansion, $S_n \geq x_n$ for every $n \geq 1$:

$$\sum_{k=0}^{n} \frac{1}{k!} \geq 1 + 1 + \frac{1}{2!}\left(1 - \frac{1}{n}\right) + \cdots + \frac{1}{n!}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\cdots\left(1 - \frac{n-1}{n}\right).$$

For every fixed $k \geq 1$, truncating the binomial expansion of x_n at its kth term and then sending $n \to \infty$ gives

$$e = \lim_{n\to\infty} x_n \geq 1 + 1 + \frac{1}{2!} + \cdots + \frac{1}{k!} = S_k.$$

By the squeeze theorem,

$$e \geq \sum_{k=0}^{\infty} \frac{1}{k!} \geq x_k \xrightarrow[k\to\infty]{} e,$$

and hence

$$\sum_{k=0}^{\infty} \frac{1}{k!} = e.$$

Note that

$$\begin{aligned}
0 < e - S_n &= \sum_{k=0}^{\infty} \frac{1}{k!} - S_n \\
&\leq \frac{1}{(n+1)!}\left(1 + \frac{1}{n+2} + \left(\frac{1}{n+2}\right)^2 + \cdots\right) \\
&= \frac{1}{(n+1)!} \cdot \frac{1}{1 - \frac{1}{n+2}} \\
&= \frac{n+2}{(n+1)!} \cdot \frac{1}{n+1} \\
&= \frac{n^2 + 2n}{n!\,(n+1)^2\, n} \leq \frac{1}{n!\, n}.
\end{aligned}$$

□

Corollary

e is irrational.

Proof

If

$$e = \frac{p}{q}, \qquad (p, q) = 1,$$

then

$$0 < e - S_q < \frac{1}{q!\, q}.$$

Thus

$$0 < q!(e - S_q) < \frac{1}{q}.$$

Since

$$S_q = \sum_{k=0}^{q} \frac{1}{k!}, \qquad q!S_q \in \mathbb{N},$$

and

$$q!e = q!\frac{p}{q} \in \mathbb{N},$$

it follows that $q!(e - S_q) \in \mathbb{N}$. Yet

$$0 < q!(e - S_q) < \frac{1}{q} \leq 1,$$

a contradiction. □

3.6 Convergence Tests

Theorem (Root Test)

For $\sum_{n=1}^{\infty} x_n$, let

$$\alpha = \limsup_{n\to\infty} |x_n|^{1/n}.$$

Then

$$\begin{aligned} \alpha < 1 &\implies \text{the series converges,} \\ \alpha > 1 &\implies \text{the series diverges,} \\ \alpha = 1 &\implies \text{the test is inconclusive.} \end{aligned}$$

Proof

The idea of this test, and of the ratio test, is to bound the series by a geometric series (and use the comparison test).
Since $\alpha < 1$, choose $\beta \in (\alpha, 1)$. There is $N \in \mathbb{N}$ such that, for every $n \geq N$,

$$|x_n|^{1/n} \leq \beta.$$

Hence

$$|x_n| \le \beta^n \quad \Longrightarrow \quad \sum_{n=N}^{\infty} |x_n| \le \sum_{n=N}^{\infty} \beta^n.$$

The latter converges because $\beta < 1$, so $\sum |x_n|$, and therefore $\sum x_n$, converges by comparison. □

Theorem (Ratio Test)

Assume $x_n \neq 0$ for all sufficiently large n. Given $\sum x_n$:

1. if
$$\limsup_{n\to\infty} \left|\frac{x_{n+1}}{x_n}\right| < 1,$$
then $\sum x_n$ is convergent;
2. if there exists $N \in \mathbb{N}$ such that
$$\left|\frac{x_{n+1}}{x_n}\right| \ge 1, \qquad \forall n \ge N,$$
then $\sum x_n$ diverges.

Proof

For (1), the proof is similar. For (2), it suffices to show divergence by observing that $\lim x_n \neq 0$. □

Proposition (Alternating Series Test)

Suppose

$$a_n \ge 0 \quad (\forall n \in \mathbb{N}), \qquad \{a_n\} \text{ decreases}, \qquad \lim_{n\to\infty} a_n = 0.$$

Then

$$\sum (-1)^n a_n$$

converges.

Proof

Let

$$S_n := \sum_{i=1}^{n} (-1)^i a_i.$$

Choose $N \in \mathbb{N}$ such that $a_n < \varepsilon$ for $n \ge N$. If $N \le n \le k$, the alternating character and the decrease of a_n give

$$|S_k - S_n| = \left| \sum_{i=n+1}^{k} (-1)^i a_i \right| \le a_{n+1} < \varepsilon.$$

The same estimate follows after interchanging n and k. Hence the partial sums are Cauchy and the series converges in $\mathbb{R}$. □

Theorem (Abel's Criterion)

Suppose

$$a_n \geq 0, \quad \{a_n\} \text{ is decreasing}, \quad \lim_{n\to\infty} a_n = 0.$$

Let $b_n \in \mathbb{R}$, and suppose the sequence of partial sums

$$\left\{\sum_{i=1}^{n} b_i\right\}_n$$

is bounded. Then $\sum a_n b_n$ converges.

Proof

Let $B_i = \sum_{j=1}^{i} b_j$, and choose M with $|B_i| \leq M$. Then

$$\begin{aligned}
\left|\sum_{i=n+1}^{k} a_i b_i\right| &= \left|\sum_{i=n+1}^{k} a_i(B_i - B_{i-1})\right| \\
&= \left|\sum_{i=n+1}^{k} a_i B_i - \sum_{i=n}^{k-1} a_{i+1} B_i\right| \\
&= \left|\sum_{i=n+1}^{k-1} (a_i - a_{i+1})B_i + a_k B_k - a_{n+1} B_n\right| \\
&\leq \sum_{i=n+1}^{k-1} |a_i - a_{i+1}|\,|B_i| + |a_k B_k| + |a_{n+1} B_n| \\
&\leq \sum_{i=n+1}^{k-1} |a_i - a_{i+1}| M + a_k M + a_{n+1} M \\
&= M\left(\sum_{i=n+1}^{k-1} (a_i - a_{i+1}) + a_k + a_{n+1}\right) \\
&= 2a_{n+1} M. \qquad (*)
\end{aligned}$$

Since $a_n \to 0$, equation $(*)$ shows that the partial sums of $\sum a_n b_n$ are Cauchy, hence convergent. □

4 Topology and Compactness

4.1 Topological Spaces, Bases, and Separability

Definition (Topology)

Let X be a set. A topology on a set X is a collection of subsets of X, $\mathcal{T} \subset \mathcal{P}(X)$, that satisfies the following three axioms:

1. $\varnothing, X \in \mathcal{T}$;
2. $\mathcal{T}$ is closed under arbitrary union;
3. $\mathcal{T}$ is closed under arbitrary finite intersection.

Thus $\mathcal{T}$ is the collection of open sets of X.

Example

Let $X = \mathbb{R}^2$, and define

$$d_2(x,y) := \|x-y\|_2, \qquad d_1(x,y) := \|x-y\|_1, \qquad d_\infty(x,y) := \|x-y\|_\infty.$$

For $x \in \mathbb{R}^2$, the relevant norm formulas are

$$\|x\|_1 := \sum_i |x_i|, \qquad \|x\|_p := \left(\sum_i |x_i|^p\right)^{1/p} \quad (1 \le p < \infty), \qquad \|x\|_\infty := \max_{1\le i\le 2} |x_i|.$$

The metrics d_1, d_2, d_∞ are different, but they induce the same topology on $\mathbb{R}^2$. More generally, all norms on a finite-dimensional vector space are equivalent. The topology induced by d_2 is the standard Euclidean topology.

Definition

Let X be a set and $\mathcal{T}$ be a topology on X. Then $(X, \mathcal{T})$ is a topological space. A basis of a set X is a subset
$\mathcal{B} \subset \mathcal{P}(X)$ such that:

1. $$X = \bigcup_{S \in \mathcal{B}} S;$$
2. for every $B_1, B_2 \in \mathcal{B}$ and every $x \in B_1 \cap B_2$, there exists $B \in \mathcal{B}$ such that
$$x \in B, \qquad B \subset B_1 \cap B_2.$$

Definition

Let $(X, \mathcal{T})$ be a topological space. Then

$$\boxed{\mathcal{B} \subset \mathcal{T}}$$

is a basis of $(X, \mathcal{T})$ if, for every $U \in \mathcal{T}$ and every $x \in U$, there exists $B \in \mathcal{B}$ such that

$$x \in B \subset U.$$

Remark

Because $\mathcal{B} \subset \mathcal{T}$, every basis element is open. Once a basis is fixed, the open sets are precisely the unions of basis elements.

Proposition

Let $\mathcal{T}$ be a topology on X and $\mathcal{B} \subset \mathcal{P}(X)$. Then $\mathcal{B}$ is a basis for $(X, \mathcal{T})$ if and only if $\mathcal{B}$ satisfies the two set-basis axioms above and

$$\mathcal{T} = \mathcal{T}_{\mathcal{B}},$$

where $\mathcal{T}_{\mathcal{B}}$ is the topology generated by $\mathcal{B}$, defined by

$$\mathcal{T}_{\mathcal{B}} := \left\{ \bigcup_{S \in A} S \;\middle|\; A \subset \mathcal{B} \right\}$$

In other words, $\mathcal{T}_{\mathcal{B}}$ is the collection of arbitrary unions of elements of $\mathcal{B}$.

Proof

If $\mathcal{B}$ is a basis for $(X, \mathcal{T})$, then the basis definition immediately gives the two set-basis axioms. Every $U \in \mathcal{T}$ is a union of elements of $\mathcal{B}$, so $\mathcal{T} \subset \mathcal{T}_{\mathcal{B}}$. Conversely, $\mathcal{B} \subset \mathcal{T}$, and $\mathcal{T}$ is closed under arbitrary unions, so $\mathcal{T}_{\mathcal{B}} \subset \mathcal{T}$.
Conversely, the two set-basis axioms imply that $\mathcal{T}_{\mathcal{B}}$ is a topology and that $\mathcal{B}$ is a basis for it. Therefore, if $\mathcal{T} = \mathcal{T}_{\mathcal{B}}$, then $\mathcal{B}$ is a basis for $(X, \mathcal{T})$. □

Definition (Separability and second countability)

A metric space is *separable* if it contains an at-most-countable dense subset. The topological space $(X, \mathcal{T})$ is *second countable* if $\mathcal{T}$ has an at-most-countable basis.

Theorem (Countable ball decomposition)

Let M be a separable metric space. Then every open set $X \subset M$ is a union of an at-most-countable family of balls with centers in X and positive rational radii:

$$X = \bigcup_i B_{r_i}(x_i), \qquad x_i \in X, \qquad r_i \in \mathbb{Q}_{>0}.$$

Proof

Since M is separable, there exists $\{x_n\}_n \subset M$ that is dense in M. Define

$$\mathcal{S} := \{B_r(x_i) \mid i \in \mathbb{N},\ r \in \mathbb{Q}_{>0},\ B_r(x_i) \subset X\}.$$

We claim

$$X = \bigcup_{S \in \mathcal{S}} S.$$

The family $\mathcal{S}$ is indexed by a subset of $\mathbb{N} \times \mathbb{Q}_{>0}$, so it is at most countable. It is also clear from the definition of $\mathcal{S}$ that

$$\bigcup_{S \in \mathcal{S}} S \subset X.$$

We want to show the reverse inclusion. Let $y \in X$. Since X is open, there exists $\delta > 0$ such that $B_\delta(y) \subset X$. Since $\{x_i\}_i$ is dense in M, there exists $x_n \in \{x_i\}_i$ such that

$$d_M(x_n, y) < \frac{\delta}{4}.$$

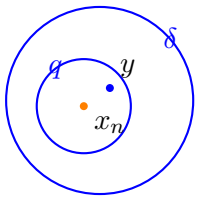
Choose $q \in \mathbb{Q}_{>0}$ such that

$$d_M(x_n, y) < q < \delta - d_M(x_n, y).$$

Then $y \in B_q(x_n)$. Moreover, for every $x \in B_q(x_n)$,

$$d_M(x, y) \leq d_M(x, x_n) + d_M(x_n, y) < q + d_M(x_n, y) < \delta,$$

so $x \in B_\delta(y)$. The containment is illustrated below:



Therefore

$$B_q(x_n) \subset B_\delta(y) \subset X \quad \Longrightarrow \quad B_q(x_n) \in \mathcal{S}.$$

Since $y \in B_q(x_n)$,

$$y \in \bigcup_{S \in \mathcal{S}} S \quad \Longrightarrow \quad X \subset \bigcup_{S \in \mathcal{S}} S \quad \Longrightarrow \quad X = \bigcup_{S \in \mathcal{S}} S.$$

□

Theorem

A metric space M is separable if and only if M is second countable.

Proof

($\Rightarrow$) Verify that

$$\mathcal{S} := \{B_r(x_i) \mid \{x_i\}_i \text{ is dense in } M,\ r \in \mathbb{Q}_{>0},\ B_r(x_i) \subset M\}$$

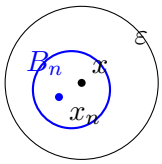
is a basis for M, by the same containment argument used in the countable ball decomposition.
($\Leftarrow$) Suppose $\mathcal{B}$ is a countable basis of the topology induced by the metric on M, say

$$\mathcal{B} = \{B_n\}_{n=1}^{\infty}.$$

After discarding the empty set if it occurs in the basis, choose $x_n \in B_n$ for every n, and let

$$D := \{x_n\}_{n=1}^{\infty}.$$

Then D is dense in M. Indeed, any ε-ball around $x \in M$ contains a basis element B_n together with its chosen point $x_n \in B_n$:



□

4.2 Closed Sets and Compactness

Proposition

$$A \subset \mathbb{R} \text{ is closed} \iff A^c \text{ is open in } \mathbb{R}.$$

Proof

Trivial. □

Properties

1. A finite intersection of open sets is open.
2. A finite union of closed sets is closed.
3. An infinite union of open sets is open.
4. An infinite intersection of closed sets is closed.

Proof

(1) and (2) are immediate. Statements (3) and (4) follow from the closed-complement characterization and De Morgan's laws. □

Definition

A set $A \subset \mathbb{R}$ satisfies the *finite open-covering property* if, for every open cover

$$\mathcal{I} = \{A_i : i \in I\}$$

of A, there is a finite subcover of A; that is, there are $i_1, \ldots, i_k$ such that

$$A \subset \bigcup_{j=1}^{k} A_{i_j}.$$

Lemma

Let $U \subset \mathbb{R}$ be open. There are $s_n \in \mathbb{Q}$, $r_n \in \mathbb{Q}_{>0}$, such that

$$U = \bigcup_{n=1}^{\infty} B_{r_n}(s_n).$$

The same construction works in any separable metric space.

Proof

Let

$$E := U \cap \mathbb{Q}, \qquad \mathcal{S} := \{B_r(x) : B_r(x) \subset U,\ x \in E,\ r \in \mathbb{Q}_{>0}\}.$$

We claim

$$U = \bigcup_{S \in \mathcal{S}} S.$$

The inclusion $\bigcup_{S\in\mathcal{S}} S \subset U$ follows from the definition of $\mathcal{S}$. Also $E \subset \bigcup_{S\in\mathcal{S}} S$. Let $x \in U \setminus E$. Then $x \in \mathbb{Q}^c \cap U$. Since U is open, there exists $\varepsilon > 0$ such that

$$B_\varepsilon(x) \subset U.$$

By density of $\mathbb{Q}$ in $\mathbb{R}$,
there exists $r \in \mathbb{Q}$ such that

$$|r - x| < \frac{\varepsilon}{4}.$$

By density of $\mathbb{Q}$, choose $s \in \mathbb{Q}_{>0}$ such that

$$|r - x| < s < \varepsilon - |r - x|.$$

Then

$$x \in B_s(r) \subset B_\varepsilon(x) \subset U,$$

so $B_s(r) \in \mathcal{S}$. Thus

$$x \in \bigcup_{S \in \mathcal{S}} S.$$

□

Lemma

Let $A \subset \mathbb{R}$. Every open cover of A has an at-most-countable subcover.

Proof

Let I be the index set, and let $\{A_i : i \in I\}$ be the open cover of A. Since every A_i is open, for each $i \in I$ there are

$$\{r_{in}\}_{n=1}^{\infty} \subset \mathbb{Q}_{>0}, \qquad \{s_{in}\}_{n=1}^{\infty} \subset \mathbb{Q},$$

such that

$$A_i = \bigcup_{n=1}^{\infty} B_{r_{in}}(s_{in}). \qquad (*)$$

Therefore

$$\bigcup_{i\in I} A_i = \bigcup_{i\in I}\bigcup_{n=1}^{\infty} B_{r_{in}}(s_{in}).$$

Since every pair (r_{in}, s_{in}) lies in $\mathbb{Q}\times\mathbb{Q}$, this union is at most countable.
Let

$$\bigcup_{i\in I} A_i = S.$$

By re-enumeration,

$$S = \bigcup_{j=1}^{\infty} B_{r_j}(s_j).$$

For every $j\in\mathbb{N}$, choose $A_j \in \{A_i : i\in I\}$ such that

$$B_{r_j}(s_j) \subset A_j;$$

such an A_j exists because of $(*)$. Hence

$$S = \bigcup_j B_{r_j}(s_j) \subset \bigcup_j A_j,$$

and $\{A_j\}$ is an at-most-countable subcover. □

Theorem

$A\subset\mathbb{R}$ is sequentially compact $\iff$ A is compact.

Proof

For $(\Rightarrow)$, suppose $A\subset\mathbb{R}$ is sequentially compact. Let $\{A_\alpha : \alpha\in I\}$ be an open cover of A. By the countable-subcover lemma, assume the index set is at most countable. Suppose there is no finite subcover, and write the cover as

$$\{A_n : n\in\mathbb{N}\}.$$

Since no finite subfamily covers A, choose

$$x_n \in A \setminus \bigcup_{i=1}^{n} A_i.$$

By sequential compactness there is a subsequence $\{x_{n_k}\}_k \subset \{x_n\}$ such that

$$x_{n_k} \xrightarrow[k\to\infty]{} x \in A.$$

Since the A_n's cover A, $x\in A_N$ for some $N\in\mathbb{N}$. Because A_N is open, there is $\varepsilon>0$ with

$$B_\varepsilon(x) \subset A_N.$$

Since $x_{n_k}\to x$,
there is $M\in\mathbb{N}$ such that, for every $k\geq M$,

$$|x_{n_k} - x| < \varepsilon \quad\Longrightarrow\quad x_{n_k}\in B(x,\varepsilon).$$

Choose $k \geq \max\{M, N\}$. Since $n_k \geq k \geq N$, the construction of $\{x_n\}$ gives

$$x_{n_k} \notin A_N,$$

whereas convergence gives

$$x_{n_k} \in B_\varepsilon(x) \subset A_N,$$

a contradiction. Thus a finite subcover exists.
For ($\Leftarrow$), suppose A is compact. We show that A is closed and bounded.
Boundedness. Consider the open cover

$$\{B_2(x) : x \in A\}.$$

It has a finite subcover

$$\{B_2(x_i) : i = 1, 2, \ldots, n\},$$

and it follows that A is bounded.
Closedness. Let $x \in A^c$. We claim that x is an interior point of A^c. The family

$$\left\{\mathbb{R} \setminus B_{1/n}[x] : n \in \mathbb{N}\right\}$$

is an open cover of A. A finite subcover, together with the nested nature of these open sets, gives an $N \in \mathbb{N}$ such that

$$A \subset \mathbb{R} \setminus B_{1/N}(x).$$

Therefore

$$B_{1/(N+1)}(x) \cap A = \varnothing, \qquad B_{1/(N+1)}(x) \subset A^c.$$

Thus x is an interior point of A^c, so A^c is open and A is closed. □

Proposition (Local boundedness on a compact set)

Suppose $A \subset \mathbb{R}$, $A \neq \varnothing$, is compact. Let $f : A \to \mathbb{R}$ be locally bounded; that is, for every $x \in A$, there exist $\delta(x) > 0$, $M(x) > 0$ such that

$$|c - x| \leq \delta(x), \quad c \in A \quad \Longrightarrow \quad |f(c)| \leq M(x).$$

Then $f(A)$ is bounded; that is, there exists $M > 0$ such that

$$|f(x)| \leq M, \qquad \forall x \in A.$$

Proof

The family

$$\{B_{\delta(x)}(x) \mid x \in A\}$$

is an open cover of A. Since A is compact, there exist $x_1, \ldots, x_n \in A$ such that

$$A \subset \bigcup_{i=1}^{n} B_{\delta(x_i)}(x_i).$$

Choose

$$M := \max_{1 \le i \le n} M(x_i).$$

Then $|f(x)| \le M$ for every $x \in A$. $\square$

4.3 Cantor Set

Definition

A set $A \subset \mathbb{R}$ is *perfect* if it is closed and every point of A is a limit point of A. Equivalently, $A = A'$, where A' denotes the set of limit points of A.

Definition (Cantor set)

Construct sets as follows. Let

$$A_0 = [0, 1].$$

Remove the open interval in the middle:

$$A_1 = A_0 \setminus \left(\frac{1}{3}, \frac{2}{3}\right) = [0, \tfrac{1}{3}] \cup [\tfrac{2}{3}, 1].$$

Then, from each interval in the union, remove the open middle third:

$$A_2 = [0, \tfrac{1}{9}] \cup [\tfrac{2}{9}, \tfrac{1}{3}] \cup [\tfrac{2}{3}, \tfrac{7}{9}] \cup [\tfrac{8}{9}, 1],$$

$$\begin{aligned} A_3 = [0, \tfrac{1}{27}] \cup [\tfrac{2}{27}, \tfrac{3}{27}] \cup [\tfrac{6}{27}, \tfrac{7}{27}] \cup [\tfrac{8}{27}, \tfrac{9}{27}] \\ \cup [\tfrac{18}{27}, \tfrac{19}{27}] \cup [\tfrac{20}{27}, \tfrac{21}{27}] \cup [\tfrac{24}{27}, \tfrac{25}{27}] \cup [\tfrac{26}{27}, 1], \end{aligned}$$

and so on.
The sequence $\{A_n\}_{n=1}^{\infty}$ consists of compact sets (closed and bounded), with

1. $$\cdots \subset A_{n+1} \subset A_n \subset \cdots \subset A_2 \subset A_1 \subset A_0;$$

2. A_n is the union of 2^n intervals, each of length 3^{-n}.

Define

$$E := \bigcap_{n=1}^{\infty} A_n$$

to be the Cantor set.

Proposition

$$E \neq \varnothing.$$

Proof

By the nested-interval property of $\mathbb{R}$: each A_n contains a nested interval from A_0; for example,

$$\cdots \subset [\tfrac{24}{27}, \tfrac{25}{27}] \subset [\tfrac{8}{9}, 1] \subset [\tfrac{2}{3}, 1] \subset [0, 1].$$

Therefore

$$E = \bigcap_{n=1}^{\infty} A_n \neq \varnothing.$$

□

Proposition

1. E does not contain any interval.
2. The Cantor set has Lebesgue measure zero.

Proof

For (1), let $(a, b) \subset [0, 1]$. Choose $m \in \mathbb{N}$ such that

$$2 \cdot 3^{-m} < b - a.$$

There is an integer k, $0 \leq k < 3^m$, such that

$$[k3^{-m}, (k+1)3^{-m}] \subset (a, b),$$

and its open middle third is

$$\left(\frac{3k+1}{3^{m+1}}, \frac{3k+2}{3^{m+1}}\right) \subset [k3^{-m}, (k+1)3^{-m}] \subset (a, b).$$

Every point in this middle third has a ternary digit equal to 1, so the open interval is disjoint from E. Consequently,

$$(a, b) \not\subset E \qquad \text{for every } (a, b) \subset [0, 1].$$

For (2), $E \subset A_n$, and A_n is a union of 2^n intervals, each of length 3^{-n}. Hence its total length is

$$m^*(E) \leq 2^n 3^{-n} = \left(\frac{2}{3}\right)^n,$$

where m^* is Lebesgue outer measure. Letting $n \to \infty$ gives $m^*(E) = 0$, and therefore E is Lebesgue measurable with $m(E) = 0$. □

Remark (Generalized Cantor sets)

By construction, the elements of E include the endpoints of the subintervals of A_n, $n \in \mathbb{N}$. The set E is called the ternary Cantor set. More generally, one can repeat the construction by removing the middle $1/p$ open interval at every step; the resulting set E_p is called the base-p Cantor set.
The space E_p is homeomorphic to the p-adic integers $\mathbb{Z}_p$.

Theorem

E is perfect.

Proof

Notice that, for every $n \in \mathbb{N}$, the endpoints of the subintervals of A_n lie in E. Moreover, every $x \in E$ lies in a chain of nested closed intervals

$$\bigcap_{n=1}^{\infty} A_{n,j(n)},$$

where
$A_{n,j(n)}$ is a subinterval of A_n:

$$\cdots \subset A_{n+1,j(n+1)} \subset A_{n,j(n)} \subset \cdots .$$

Thus the distance from x to an endpoint (which belongs to the Cantor set) can be made infinitely small, so x is a limit point of E. □

Theorem

Every nonempty perfect set S is uncountable.

Proof

Because S is perfect, it is closed in $\mathbb{R}$, and hence complete with the induced metric. Suppose, toward a contradiction, that

$$S = \{x_n : n \in \mathbb{N}\} = \bigcup_{n=1}^{\infty} \{x_n\}.$$

Each singleton $\{x_n\}$ is closed in S. It has empty interior in S, since otherwise x_n would be isolated in S, contradicting perfectness. Thus every $\{x_n\}$ is nowhere dense in the complete metric space S. The Baire Category Theorem says that a nonempty complete metric space cannot be a countable union of nowhere dense closed sets, a contradiction. Therefore S is uncountable. □

Example (Polynomial versus geometric decay)

$$\frac{n^k}{(1+d)^n} = \frac{n^k}{1 + nd + \binom{n}{2}d^2 + \binom{n}{3}d^3 + \cdots}, \qquad n^k r^n \xrightarrow[n\to\infty]{} 0 \quad (|r| < 1).$$

5 Limits and Continuity

5.1 Limits of Functions

Lemma

If x is a limit point of $A \subset \mathbb{R}$, then

$$(x-\varepsilon, x+\varepsilon) \cap A \setminus \{x\}$$

is infinite for every $\varepsilon > 0$.

Proof

Trivial. □

Definition

Let $f : A \to \mathbb{R}$ be a function, and let c be a limit point of A. The point c need not belong to A. We say

$$\lim_{x \to c} f(x) = L$$

if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < |x - c| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

Theorem (Sequential characterization of function limits)

Let c be a limit point of A, and let $f : A \to \mathbb{R}$. Then

$$\lim_{x \to c} f(x) = L \quad \Longleftrightarrow \quad \text{for every } \{x_n\} \subset A \text{ with } x_n \to c,\ f(x_n) \to L,$$

where $x_n \neq c$ if $c \in A$.

Proof

The forward implication is trivial. For the reverse implication, show that

$$\lim_{x \to c} f(x) \neq L \quad \Longrightarrow \quad \exists \{x_n\} \text{ such that } x_n \to c \text{ but } f(x_n) \not\to L.$$

□

Proposition

The limit of a function approaching a point is unique.

Proof

Suppose

$$\lim_{x \to c} f(x) = L, \qquad \lim_{x \to c} f(x) = M.$$

Show that $L = M$. □

Proposition

Suppose

$$\lim_{x\to c} f(x) = L, \qquad \lim_{x\to c} g(x) = M.$$

Then

$$\begin{aligned} \lim_{x\to c}(Kf)(x) &= KL, && \forall K \in \mathbb{R}, \\ \lim_{x\to c}(f \pm g)(x) &= L \pm M, \\ \lim_{x\to c}(fg)(x) &= LM. \end{aligned}$$

If $M \neq 0$, then

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

Proof

Use the sequential characterization. □

Theorem (Squeeze Theorem)

Let $f, g, h : A \to \mathbb{R}$, let c be a limit point of A, and suppose

$$f \leq h \leq g$$

near c. If

$$\lim_{x\to c} f(x) = L, \qquad \lim_{x\to c} g(x) = L,$$

then

$$\lim_{x\to c} h(x) = L.$$

Proof

Use the sequential characterization. □

Definition

Let $f : A \to \mathbb{R}$, and let $c \in \mathbb{R}$ be a limit point of A. We say

$$\lim_{x\to c^+} f(x) = L =: f(c^+)$$

if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < x - c < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

We say

$$\lim_{x\to c^-} f(x) = L =: f(c^-)$$

if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 < c - x < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

Finally,

$$\lim_{x\to c} f(x) = +\infty$$

if, for every $N \in \mathbb{N}$, there exists $\delta > 0$ such that

$$0 < |x - c| < \delta \quad \Longrightarrow \quad f(x) \geq N.$$

Let

$$f : (a, \infty) \to \mathbb{R}.$$

We say

$$\lim_{x\to\infty} f(x) = L$$

if, for every $\varepsilon > 0$, there is $N \in \mathbb{N}$ such that

$$x \geq N \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

The definition at $-\infty$ is analogous.

5.2 Continuous Functions

Definition

Let $f : A \to \mathbb{R}$, and let $c \in A$ (it need not be a limit point). Then f is *continuous at* $x = c$ if, for every $\varepsilon > 0$, there is $\delta > 0$ such that

$$|x - c| < \delta \quad \Longrightarrow \quad |f(x) - f(c)| < \varepsilon.$$

Theorem

- If $c \in A$ is a limit point, then

$$f \text{ is continuous at } c \quad \Longleftrightarrow \quad \lim_{x\to c} f(x) = f(c).$$

- If c is not a limit point of A, then f is continuous at c.
- Sequential characterization: f is continuous at $x = c$ if and only if, for every $\{x_n\} \subset A$ such that $x_n \to c$,

$$f(x_n) \longrightarrow f(c).$$

Theorem

Let $c \in A$ and let $f, g : A \to \mathbb{R}$. Suppose f and g are continuous at c. Then

$$kf, \qquad fg, \qquad f \pm g, \qquad \frac{f}{g}$$

are continuous at c, where $k \in \mathbb{R}$, and $g(c) \neq 0$ for f/g.

Theorem

Let $c \in A$, $f : A \to B$, and $g : B \to \mathbb{R}$. If f is continuous at $c \in A$ and g is continuous at $f(c) \in B$, then

$$g \circ f$$

is continuous at $c \in A$.

Continuous solutions of Cauchy's functional equation

Suppose $f : \mathbb{R} \to \mathbb{R}$ is continuous, with

$$f(x+y) = f(x) + f(y), \qquad \forall x, y \in \mathbb{R}.$$

Then there exists $c \in \mathbb{R}$ such that $f(x) = cx$ for every $x \in \mathbb{R}$.

Proof

$$f(0+0) = f(0) = f(0) + f(0) \quad \Longrightarrow \quad f(0) = 0.$$

Also,

$$0 = f(x-x) = f(x) + f(-x), \qquad \text{so} \qquad f(-x) = -f(x).$$

Moreover,

$$f(n) = nf(1) \quad \Longrightarrow \quad f\left(\frac{1}{n}\right) = \frac{f(1)}{n} \quad \Longrightarrow \quad f\left(\frac{p}{q}\right) = \frac{p}{q} f(1).$$

Thus $f(r) = rf(1)$ for every $r \in \mathbb{Q}$. Fix an arbitrary $x \in \mathbb{R}$. By density of $\mathbb{Q}$, choose $r_n \in \mathbb{Q}$ with $r_n \to x$. Then

$$f(x) = \lim_{n\to\infty} f(r_n) = \lim_{n\to\infty} r_n f(1) = xf(1).$$

Therefore $f(x) = cx$ for every $x \in \mathbb{R}$, with $c = f(1)$. □

Example (Thomae's function)

Let $I = (0,1)$, and define $f : I \to \mathbb{R}$ by

$$f(x) = \begin{cases} \dfrac{1}{q}, & x = \dfrac{p}{q},\ (p,q) = 1,\ p, q \in \mathbb{N}, \\ 0, & x \notin \mathbb{Q} \cap I. \end{cases}$$

Then f is continuous at every $x \in \mathbb{Q}^c \cap I$ and discontinuous at every $x \in \mathbb{Q} \cap I$.

Proof

As $x \to c \in \mathbb{Q}^c \cap I$, any rational p/q close to c must have $q \to \infty$. □

5.3 Extension Theorems

Theorem (Continuous extension from a closed subset of $\mathbb{R}$)

Suppose $A \subset \mathbb{R}$, $A \neq \varnothing$, is closed, and $f : A \to \mathbb{R}$ is continuous. Then there exists a continuous function

$$\overline{f} : \mathbb{R} \to \mathbb{R}$$

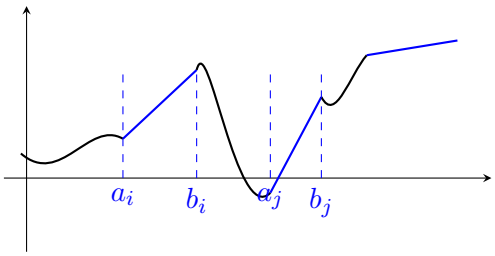
such that $\overline{f}|_A = f$.

Proof

Since A is closed, $\mathbb{R} \setminus A$ is a disjoint union of its connected components, each of which is an open interval. A bounded component has the form (a_i, b_i), with $a_i, b_i \in A$. There may also be one left-unbounded component $(-\infty, b_i)$ and one right-unbounded component (a_i, ∞).
Define $\overline{f} : \mathbb{R} \to \mathbb{R}$ by $\overline{f} = f$ on A. On each bounded component (a_i, b_i), interpolate linearly:

$$\overline{f}(x) := \frac{b_i - x}{b_i - a_i} f(a_i) + \frac{x - a_i}{b_i - a_i} f(b_i).$$

On a component $(-\infty, b_i)$, set $\overline{f}(x) := f(b_i)$; on a component (a_i, ∞), set $\overline{f}(x) := f(a_i)$. The construction interpolates linearly between endpoint values across each bounded component (a_i, b_i) of the complement:



The definition is unambiguous because the components are disjoint. It is continuous within every component and agrees with f at every finite endpoint. At an accumulation point of components, their endpoints converge to the same point of A; continuity of f, together with the fact that the interpolated value is a convex combination of the endpoint values, proves continuity there. Thus $\overline{f}$ is continuous on $\mathbb{R}$. □

Definition (Normal space)

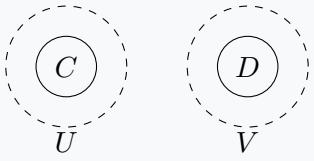
The space $(X, \mathcal{T})$ is normal if, for every closed $C, D \subset X$ such that

$$C \cap D = \varnothing, \qquad C, D \neq \varnothing,$$

there exist open U, V such that

$$U \cap V = \varnothing, \qquad C \subset U, \qquad D \subset V.$$

The separation condition is illustrated below:



Theorem (Tietze extension theorem)

Let X be a normal metric space, and let $f : A \to \mathbb{R}$ be continuous on a closed set $A \subset X$, to $\mathbb{R}$ with the standard topology. Then there exists a continuous extension of f on X, that is,

$$\exists \overline{f} \in C(X, \mathbb{R}) \quad \text{such that} \quad \overline{f}|_A = f, \qquad \inf \overline{f}(X) = \inf f(A), \qquad \sup \overline{f}(X) = \sup f(A).$$

A proof uses the separation axioms and Urysohn's lemma and is omitted here.

5.4 Continuity over Compact Sets

Lemma

Let $A \subset \mathbb{R}$ be compact, and let $f : A \to \mathbb{R}$ be continuous. Then

$$f(A) \subset \mathbb{R}$$

is compact.

Proof

One proof uses the fact that inverse images of open sets under continuous maps are open, together with the finite-open-cover characterization of compactness.
Here is a sequential proof that $f(A)$ is closed and bounded.
To show $f(A)$ is closed, let $L \in \mathbb{R}$ be a limit point of $f(A)$. Then there is a sequence

$$\{f(x_n)\}_{n=1}^{\infty} \longrightarrow L.$$

Since $\{x_n\} \subset A$ and A is compact, there is a subsequence

$$x_{n_k} \longrightarrow x \in A.$$

By continuity, $f(x) = L$.
To show that $f(A)$ is bounded, suppose otherwise. Then there is a sequence $\{f(x_n)\}_n$ such that

$$|f(x_n)| \geq n, \qquad \forall n.$$

By compactness of A, there is a subsequence

$$x_{n_k} \longrightarrow x.$$

But then there exists $N \in \mathbb{N}$ such that, for all $k \geq N$,

$$|f(x_{n_k})| \leq |f(x)| + 1,$$

a contradiction. □

Theorem (Extreme Value Theorem)

A continuous function on a nonempty compact set attains its minimum and maximum, and is bounded.

Proof

By the compact-image lemma, a continuous $f : A \to \mathbb{R}$, with A compact, has compact image

$$f(A) \subset \mathbb{R}.$$

Hence it has a least-upper-bound and greatest-lower-bound, and

$$\sup f(A),\ \inf f(A) \in f(A).$$

Thus the maximum and minimum are attained. □

Theorem (Intermediate Value Theorem)

Let $f : [a, b] \to \mathbb{R}$ be continuous, let $c < d$ lie in $[a, b]$, and suppose

$$k \in (f(c), f(d)), \qquad \{f(c), f(d)\} \subset f([a, b]).$$

Then there exists $\xi \in (c, d)$ such that

$$f(\xi) = k.$$

Proof

Bisection proof. First treat the case $k = 0$, $f(c) < 0$, and $f(d) > 0$, and then shift the function by k.
Connected-space formulation. More generally, let (X, d) be a connected metric space and let $f : X \to \mathbb{R}$ be continuous. If $a, b \in f(X)$ and $a < c < b$, then $c \in f(X)$.
Indeed, if $c \notin f(X)$, then

$$X = f^{-1}((-\infty, c)) \cup f^{-1}((c, \infty)).$$

Since f is continuous, these inverse images are open, disjoint, and nonempty, a contradiction. Similarly, if $f : X \to Y$ is continuous between metric spaces and X is connected, then $f(X)$ is connected in Y.
Supremum proof. Let

$$A := \{x \in [c, d] : f(x) < k\}.$$

Since $k \in (f(c), f(d))$, we have $c \in A$. Thus $A \neq \varnothing$ and is bounded, so

$$x_0 := \sup A$$

exists in $[c, d]$. Continuity at c, together with $f(c) < k$, gives points of A strictly to the right of c, so $x_0 > c$. Continuity at d, together with $f(d) > k$, gives a left-neighborhood of d that is disjoint from A, so $x_0 < d$. Hence

$$x_0 \in (c, d).$$

By the definition of the supremum, there is a sequence $x_n \in A$ with $x_n \to x_0$. Thus continuity gives

$$f(x_0) = \lim_{n\to\infty} f(x_n) \leq k.$$

On the other hand, every $y \in (x_0, d]$ lies outside A, so $f(y) \geq k$. Letting $y \to x_0^+$ and using continuity gives $f(x_0) \geq k$. Therefore $f(x_0) = k$. □

Continuous functions with irrational values

Suppose $f : (a, b) \to \mathbb{R}$ is continuous. If

$$f((a, b)) \subset \mathbb{R} \setminus \mathbb{Q},$$

then f is constant.

Proof

If f were nonconstant, the Intermediate Value Theorem and density of $\mathbb{Q}$ in $\mathbb{R}$ would give $c \in (a, b)$ such that $f(c) \in \mathbb{Q}$, a contradiction. □

5.5 Uniform Continuity

Definition

A function $f : A \to \mathbb{R}$ is *uniformly continuous* on A if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that, for all $x, y \in A$,

$$|x - y| < \delta \quad \Longrightarrow \quad |f(x) - f(y)| < \varepsilon.$$

Composition preserves uniform continuity

If $f : A \to B$, $g : B \to \mathbb{R}$, with $A, B \subset \mathbb{R}$, and f, g are uniformly continuous on their domains, then $g \circ f$ is uniformly continuous on A. This follows by applying the two uniform continuity estimates successively.

Products of bounded uniformly continuous functions

If $f, g : A \to \mathbb{R}$ are uniformly continuous and bounded on A, then $h = fg$ is uniformly continuous on A.

Proof

Since f, g are bounded, there exists $M > 0$ such that $|f| \leq M$, $|g| \leq M$. Let $\varepsilon > 0$. Uniform continuity gives $\delta > 0$ such that $|x - y| < \delta$ implies

$$|f(x) - f(y)| < \frac{\varepsilon}{2M}, \qquad |g(x) - g(y)| < \frac{\varepsilon}{2M}.$$

Then

$$\begin{aligned} |f(x)g(x) - f(y)g(y)| &\leq |f(x)|\,|g(x) - g(y)| + |g(y)|\,|f(x) - f(y)| \\ &< \varepsilon. \end{aligned}$$

□

Theorem (Heine–Cantor)

Let $f : A \to \mathbb{R}$, where A is compact. Then

$$f \text{ is continuous} \iff f \text{ is uniformly continuous.}$$

Proof

The reverse implication is automatic. For the forward implication, suppose otherwise: f is continuous but not uniformly continuous. By definition there is $\varepsilon_0 > 0$ such that, for every $1/n$, there exist $x_n, y_n \in A$ with

$$|x_n - y_n| < \frac{1}{n}, \qquad |f(x_n) - f(y_n)| \geq \varepsilon_0.$$

By compactness, there is a subsequence $\{x_{n_k}\} \subset \{x_n\}$ such that

$$x_{n_k} \longrightarrow x.$$

By compactness again,
there is a subsequence

$$\{y_{n_{k_\ell}}\}_\ell \subset \{y_{n_k}\}_k \subset \{y_n\}_n$$

such that

$$y_{n_{k_\ell}} \longrightarrow y.$$

Notice that $\{x_{n_{k_\ell}}\} \subset \{x_{n_k}\}$ also converges to $x \in A$, since it is a subsequence of a convergent sequence. Since

$$|x_n - y_n| \leq \frac{1}{n},$$

we have

$$\lim_{\ell \to \infty} x_{n_{k_\ell}} = x = \lim_{\ell \to \infty} y_{n_{k_\ell}} = y.$$

By continuity,

$$f(x) = \lim_{\ell \to \infty} f(x_{n_{k_\ell}}) = \lim_{\ell \to \infty} f(y_{n_{k_\ell}}).$$

This contradicts

$$|f(x_n) - f(y_n)| \geq \varepsilon_0, \qquad \forall n \in \mathbb{N}.$$

□

Example

$$f(x) = x^2$$

is continuous on $\mathbb{R}$ but not uniformly continuous on $\mathbb{R}$. It is uniformly continuous on $[0, 1]$.

5.6 Types of Discontinuities

Definition

Let $f : (a, b) \to \mathbb{R}$ be a function that is not continuous (is discontinuous) at $c \in (a, b)$. If

$$f(c^+) = \lim_{x \to c^+} f(x) \qquad \text{and} \qquad f(c^-) = \lim_{x \to c^-} f(x)$$

both exist, then f has a *discontinuity of the first kind* at c. Otherwise f has a *discontinuity of the second kind* at c.

Example

$$f(x) = \begin{cases} 1, & x \geq 0, \\ 0, & x < 0 \end{cases}$$

has a discontinuity of the first kind at $x = 0$.

Example

$$f(x) = \begin{cases} x, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q} \end{cases}$$

is continuous at 0 and has a discontinuity of the second kind at every real number other than 0.

5.7 Monotone and Inverse Functions

Definition

A function $f : [a, b] \to \mathbb{R}$ is *monotone increasing* if

$$f(x) \leq f(y), \qquad \forall x \leq y, \quad x, y \in [a, b].$$

It is *strictly (monotone) increasing* if

$$f(x) < f(y), \qquad \forall x < y, \quad x, y \in [a, b].$$

Similarly, we have monotone decreasing and strictly monotone decreasing functions.

Endpoint extrema of an increasing function

Suppose $f : [a, b] \to \mathbb{R}$ is increasing and continuous. Then

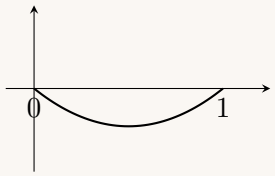
$$f(a) = \inf_{[a,b]} f, \qquad f(b) = \sup_{[a,b]} f.$$

Products need not preserve monotonicity

The functions $f(x) = x$, $g(x) = x - 1$ are increasing on $[0, 1]$, but

$$f(x)g(x) = x(x - 1)$$

is not. Its graph is:



Proposition

If f is monotone increasing on $[a,b]$, then $f(x^+)$ and $f(x^-)$ exist for every $x \in (a,b)$. Moreover,

$$\sup_{a\le t<x} f(t) = f(x^-) \le f(x) \le f(x^+) = \inf_{x<t\le b} f(t),$$

and

$$f(x^+) \le f(y^-), \qquad \forall\, a < x < y < b.$$

Proof

The set

$$\{f(t) : a \le t < x\}$$

is bounded above by $f(x)$, since f is monotone increasing. Thus

$$\sup_{a\le t<x} f(t)$$

exists. It is then trivial to prove that

$$\sup_{a\le t<x} f(t) = f(x^-)$$

using the characterization of the supremum.

$$\sup \quad \Longleftrightarrow \quad \forall \varepsilon > 0,\ \exists x \text{ such that } \sup -\varepsilon < x \le \sup .$$

Likewise we prove this for $\inf_{x\in E} f(x)$.

Proposition (Bounded monotone functions are uniformly continuous)

If $f : (a,b) \to \mathbb{R}$ is continuous, bounded, and increasing, then f is uniformly continuous on (a,b).

Proof

Since f is increasing and bounded,

$$\lim_{x\to a^+} f(x) = \inf f((a,b)), \qquad \lim_{x\to b^-} f(x) = \sup f((a,b)).$$

Defining $\widetilde{f}(a)$ and $\widetilde{f}(b)$ by these one-sided limits gives a continuous extension $\widetilde{f} : [a,b] \to \mathbb{R}$. The Heine–Cantor theorem makes $\widetilde{f}$, and hence its restriction f, uniformly continuous. □

Continuity and boundedness do not suffice

The function

$$f(x) = \sin\left(\frac{1}{x}\right)$$

is bounded and continuous, but not increasing and not uniformly continuous on $(0, 1)$.

Corollary

All discontinuity points of monotonic functions are of first kind.

Proposition

If f is monotone on (a, b), then the set of discontinuities of f is at most countable.

Proof

We show there is an injective map to $\mathbb{Q}$. Without loss of generality, suppose f is monotone increasing. Then by the previous proposition, $f(x-)$ and $f(x+)$ exist. Suppose

$$E := \{x \in (a, b) \mid f \text{ is discontinuous at } x\}, \qquad x \in E.$$

Notice that $f(x-) \neq f(x+)$. If otherwise,

$$f(x-) = f(x) = f(x+)$$

by the previous proposition (since f is monotone increasing), hence f is continuous at x. Thus $f(x-) < f(x+)$. By density of $\mathbb{Q}$ in $\mathbb{R}$, there exists $r_x \in \mathbb{Q}$ such that

$$r_x \in (f(x-), f(x+)).$$

In this way we can define a map

$$\varphi : E \longrightarrow \mathbb{Q}, \qquad x \longmapsto r_x.$$

We claim it is injective.
Suppose there exist $x, y \in E$ (discontinuities of f) such that $r_x = r_y$. Suppose $x \neq y$. Then either $x < y$ or $x > y$. By the previous proposition,

$$x < y: \quad f(x-) \leq f(x+) \leq f(y-) \leq f(y+),$$
$$x > y: \quad f(x-) \geq f(y+).$$

The two possible orderings are:



Yet, since $r_x = r_y$,

$$r_x \in (f(x-), f(x+)), \qquad r_y \in (f(y-), f(y+)),$$

and hence

$$r_x = r_y \in (f(x-), f(x+)) \cap (f(y-), f(y+)) \quad \Longrightarrow \quad (f(x-), f(x+)) \cap (f(y-), f(y+)) \neq \varnothing.$$

Yet in either case $x < y$ or $y < x$, the intersection is empty. Thus $x = y$, so φ is injective, and

$$|E| \leq |\mathbb{Q}|.$$

Therefore E is at most countable. □

Continuous injective functions are strictly monotone

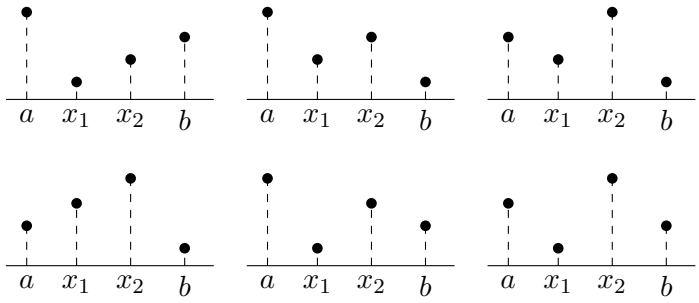
Suppose $f : [a, b] \to \mathbb{R}$ is continuous and does not take any of its values twice. Then f is strictly monotone on $[a, b]$.

Proof

Without loss of generality, suppose $f(a) > f(b)$. They are not equal since f is injective. Suppose f is not strictly monotone. Then there exist $x_1 < x_2 \in [a, b]$ such that

$$f(x_1) < f(x_2).$$

The six possible vertical orderings are illustrated below:



Equivalently, the six vertical orderings are

$$\begin{aligned}
&f(x_1) < f(x_2) < f(b) < f(a),\\
&f(b) < f(x_1) < f(x_2) < f(a),\\
&f(b) < f(x_1) < f(a) < f(x_2),\\
&f(b) < f(a) < f(x_1) < f(x_2),\\
&f(x_1) < f(b) < f(x_2) < f(a),\\
&f(x_1) < f(b) < f(a) < f(x_2).
\end{aligned}$$

Without loss of generality, consider the ordering

$$f(x_1) < f(x_2) < f(b) < f(a).$$

By the Intermediate Value Theorem, there exist $x \in (x_1, x_2)$ and $y \in (a, x_1)$ such that $f(x) = f(y)$. Thus f is not injective. If $x_1 = a$ or $x_2 = b$, the argument is simpler. The other cases are similar, leading to a contradiction. Therefore f must be strictly monotone. □

Proposition

If f is continuous and strictly increasing on $[a, b]$, then f is bijective:

$$f : [a, b] \longrightarrow [f(a), f(b)].$$

Moreover, its inverse is also continuous.

Proof

The map $f : [a, b] \to [f(a), f(b)]$ is surjective by the Intermediate Value Theorem. It is injective since f is strictly increasing. Let

$$f^{-1} : [f(a), f(b)] \longrightarrow [a, b]$$

be the inverse. We give three different ways of proving continuity.

(1) Suppose otherwise that f^{-1} is discontinuous at $y_0 \in [f(a), f(b)]$. Since f^{-1} is strictly increasing, a discontinuity creates a jump: with the evident one-sided modification at an endpoint, the inequality

$$f^{-1}(y_0-) < f^{-1}(y_0+)$$

shows that the open interval between these two one-sided limits contains points outside the range of f^{-1}. This contradicts the surjectivity of f^{-1} onto $[a, b]$.

(2) Suppose $\{y_n\}_{n=1}^{\infty} \subset [f(a), f(b)]$ converges to $y \in [f(a), f(b)]$. Set

$$x_n := f^{-1}(y_n), \qquad x_0 := f^{-1}(y), \qquad f(x_n) = y_n \longrightarrow y = f(x_0).$$

Suppose $x_n \not\to x_0$. Then some subsequence remains a fixed positive distance from x_0. Compactness of $[a, b]$ gives a further subsequence, still denoted x_{n_k}, such that

$$x_{n_k} \longrightarrow x, \qquad x \neq x_0.$$

Continuity of f gives

$$f(x) = \lim_{k\to\infty} f(x_{n_k}) = y = f(x_0),$$

contradicting injectivity. Hence $x_n \to x_0$, proving sequential continuity of f^{-1}.

(3) The previous methods both used continuity of f, but it is not needed. Let $\varepsilon > 0$. We seek $\delta > 0$ such that

$$|y - y_0| < \delta, \quad y \in f([a, b]) \quad \Longrightarrow \quad |f^{-1}(y) - f^{-1}(y_0)| < \varepsilon.$$

Let $x_0 := f^{-1}(y_0)$, and set $\eta := \min\{\varepsilon/2, (b - a)/2\}$. Use whichever of the comparison points $x_0 - \eta$ and $x_0 + \eta$ lie in $[a, b]$. The corresponding positive gaps are

$$y_0 - f(x_0 - \eta) \quad \text{and} \quad f(x_0 + \eta) - y_0.$$

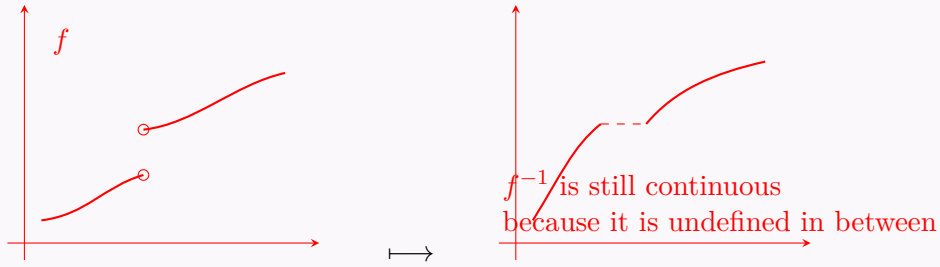
Let δ be the minimum of the available gaps. At least one comparison point is available, and strict monotonicity makes every available gap positive. If one comparison point lies outside $[a, b]$, the corresponding side of the desired estimate is automatic. Thus, for every $y \in f([a, b])$,

$$|y - y_0| < \delta \quad \Longrightarrow \quad |f^{-1}(y) - f^{-1}(y_0)| < \eta < \varepsilon.$$

□

Continuity on the image

Continuity of f is unnecessary for continuity of its inverse on the image: every strictly increasing function $f : [a, b] \to \mathbb{R}$ is bijective onto $f([a, b])$, and f^{-1} is continuous in the subspace topology on that image. The diagrams show that f may have jumps; the corresponding gaps lie outside the domain of f^{-1}.



5.8 Lipschitz and Hölder Continuity

Definition

A function $f : A \to \mathbb{R}$ is called Lipschitz (continuous) if there exists $K > 0$ such that

$$|f(x_1) - f(x_2)| \leq K|x_1 - x_2|,$$

for all $x_1, x_2 \in A$.

Proposition

If $f : A \to \mathbb{R}$ is Lipschitz, then f is uniformly continuous on A.

Proof

Trivial. □

Example

Let $\varnothing \neq A \subset \mathbb{R}$. Define

$$d_A(x) := \inf\{|x - y| : y \in A\}, \qquad \forall x \in \mathbb{R}.$$

Then $d_A : \mathbb{R} \to \mathbb{R}$ is Lipschitz continuous.

Proof

Let $x_1, x_2 \in \mathbb{R}$. For every $a \in A$, the triangle inequality gives

$$d_A(x_1) \leq |x_1 - a| \leq |x_1 - x_2| + |x_2 - a|.$$

Taking the infimum over $a \in A$ yields

$$d_A(x_1) - d_A(x_2) \leq |x_1 - x_2|.$$

Interchanging x_1 and x_2 gives

$$d_A(x_2) - d_A(x_1) \leq |x_1 - x_2|.$$

Combining the inequalities,

$$|d_A(x_2) - d_A(x_1)| \leq |x_1 - x_2|.$$

□

Uniform continuity is not preserved by unbounded products

Define $g : \mathbb{R} \to \mathbb{R}$ by

$$g(x) := 4\left(x - n - \frac{1}{2}\right)^2 - 1, \qquad x \in [n, n+1], \quad n \in \mathbb{Z}.$$

Then g is uniformly continuous on $\mathbb{R}$. Moreover, the identity function is uniformly continuous on $\mathbb{R}$, while $xg(x)$ is not.

Proof

The range of g lies in $[-1, 0]$. Hence, if $|x - y| \geq 1/8$, then

$$|g(x) - g(y)| \leq 1 \leq 8|x - y|.$$

If $|x - y| < 1/8$, then x, y lie either in the same unit interval or in adjacent unit intervals. Suppose first that $x \in [n, n+1]$ and $y \in [n+1, n+2]$. Then

$$\begin{aligned}
|g(x) - g(y)| &\leq |g(x) - g(n+1)| + |g(n+1) - g(y)| \\
&= 4|x - n|\,|x - (n+1)| + 4|y - (n+1)|\,|y - (n+2)| \\
&= 4|x - n|\,[n + 1 - x] + 4|y - (n+1)|\,[n + 2 - y] \\
&\leq 4[(n+1) - x + y - (n+1)] \\
&= 4|y - x| \leq 8|y - x|.
\end{aligned}$$

If $x, y \in [n, n+1]$,

$$\begin{aligned}
|g(x) - g(y)| &= 4|x + y - 2n - 1|\,|x - y| \\
&\leq 4(|x - n| + |y - n - 1|)|x - y| \\
&< 8|x - y|.
\end{aligned}$$

Thus g is Lipschitz on $\mathbb{R}$, hence uniformly continuous on $\mathbb{R}$. The function $f(x) = x$ is also Lipschitz and therefore uniformly continuous.
The product fg is not uniformly continuous. Choose $\varepsilon_0 = 1$. Let $\delta > 0$, and define

$$\eta := \min\left\{\delta, \frac{1}{100}\right\}.$$

Choose $n \in \mathbb{N}$ such that $n \geq 1/\eta$. Choose

$$y = n + 1, \qquad x = n + 1 - \frac{\eta}{2} = y - \frac{\eta}{2}.$$

Then $|x-y| = \eta/2 < \delta$. Compute

$$|f(x)g(x) - f(y)g(y)| = \left| x\left[4\left(x-n-\frac{1}{2}\right)^2 - 1\right] - y\left[4\left(y-n-\frac{1}{2}\right)^2 - 1\right]\right|$$

and

$$\begin{aligned}
&= \left| x\left[4\left(x-n-\frac{1}{2}\right)^2 - 1\right] - (n+1)\left[4\left(\frac{1}{2}\right)^2 - 1\right]\right| \\
&= |x|\,|2x-2n-1-1|\,|2x-2n-1+1| \\
&= |x| \cdot 2|x-n-1| \cdot 2|x-n| \\
&= |x| \cdot 4|x-y| \cdot |x-y+1| \\
&= x \cdot 4(y-x)\left(1-\frac{\eta}{2}\right) \\
&= x \cdot 4\frac{\eta}{2}\left(1-\frac{\eta}{2}\right) \geq x \cdot 4\frac{\eta}{2}\cdot\frac{1}{2} \\
&= x\eta \geq n\eta \geq 1 = \varepsilon_0, \qquad \left(x = n + \left(1-\frac{\eta}{2}\right)\right).
\end{aligned}$$

Thus fg is not uniformly continuous. □

Definition (Hölder continuity)

A function $f : A \to \mathbb{R}$ is Hölder continuous on A if there exist $K > 0$, $\alpha > 0$, such that

$$|f(x_1) - f(x_2)| \leq K|x_1 - x_2|^\alpha, \qquad \forall x_1, x_2 \in A.$$

Proposition (Hölder continuity implies uniform continuity)

If $f : A \to \mathbb{R}$ is Hölder continuous, then f is uniformly continuous on A.

The square-root function

The function

$$f : [0,\infty) \to \mathbb{R}, \qquad x \longmapsto \sqrt{x},$$

is uniformly continuous but not Lipschitz on $[0,\infty)$.

Proof

Let $\varepsilon > 0$. Choose $\delta = \varepsilon^2$. Then

$$|\sqrt{x} - \sqrt{y}|^2 = |\sqrt{x} - \sqrt{y}|\,|\sqrt{x} - \sqrt{y}| \leq |\sqrt{x} - \sqrt{y}|\,|\sqrt{x} + \sqrt{y}| = |x-y| < \varepsilon^2.$$

Therefore $|\sqrt{x} - \sqrt{y}| < \varepsilon$.
The function f is not Lipschitz. Suppose there exists $K > 0$ such that

$$|f(x) - f(y)| \leq K|x-y|, \qquad \forall x, y \in [0,\infty).$$

Choose $x = 0$. Then

$$|f(y)| \leq K|y|, \qquad \forall y \geq 0.$$

Yet for $y = 1/(K^2+1)$,

$$|f(y)| = \frac{1}{\sqrt{K^2+1}} > \frac{K}{K^2+1} = K|y|,$$

a contradiction. □

6 Differentiation

6.1 The Derivative and Inverse Functions

Definition

Let $I \subset \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$, and let $c \in I$. We say f is differentiable at c if

$$f'(c) := \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

exists. If so, we say this is the derivative of f at c.

Theorem

If $f : I \to \mathbb{R}$ has a derivative at $c \in I$, then f is continuous at c.

Proof

Since $c \in I$ is a cluster point, it suffices to show $\lim_{x\to c} f(x) = f(c)$. Notice, for every $x \neq c$,

$$f(x) - f(c) = \frac{f(x) - f(c)}{x - c}(x - c).$$

Hence

$$\lim_{x\to c}(f(x) - f(c)) = \lim_{x\to c}\left\{\frac{f(x) - f(c)}{x - c}(x - c)\right\} = f'(c)\lim_{x\to c}(x - c) = 0,$$

so $\lim_{x\to c} f(x) = f(c)$. □

Example

$y = |x|$ is continuous everywhere but not differentiable at $c = 0$.

Theorem

Let $I \subset \mathbb{R}$ be an interval, $c \in I$, and let $f, g : I \to \mathbb{R}$ be differentiable at c. Then the following functions are differentiable, and

$$\begin{aligned}(kf)'(c) &= kf'(c) && (k \in \mathbb{R}),\\ (f+g)'(c) &= f'(c) + g'(c),\\ (fg)'(c) &= f'(c)g(c) + f(c)g'(c).\end{aligned}$$

If $g(c) \neq 0$, then

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Definition

The function f is differentiable on $I \subset \mathbb{R}$ if f is differentiable for every $c \in I$. This gives a new function $f' : I \to \mathbb{R}$.

Proposition (Integer power rule)

For $f(x) = x^n$, $n \in \mathbb{N}_{>0}$,

$$f'(x) = nx^{n-1}.$$

Proof

By the binomial theorem, for $h \neq 0$,

$$\frac{(x+h)^n - x^n}{h} = \sum_{k=1}^{n} \binom{n}{k} x^{n-k} h^{k-1}.$$

Letting $h \to 0$, every term with $k \geq 2$ vanishes, while the term with $k = 1$ tends to nx^{n-1}. Thus $f'(x) = nx^{n-1}$. □

Theorem (Carathéodory's theorem)

Let $I \subset \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$, and $c \in I$. Then f is differentiable at c if and only if there exists a function $\varepsilon : I \to \mathbb{R}$, continuous at c, such that

$$f(x) = f(c) + \varepsilon(x)(x - c), \qquad \forall x \in I.$$

Moreover, $f'(c) = \varepsilon(c)$ in this case.
For the forward implication, define

$$\varepsilon(x) = \begin{cases} \dfrac{f(x) - f(c)}{x - c}, & x \neq c, \\ f'(c), & x = c, \end{cases}$$

Differentiability of f at c makes ε continuous at c.

Proof

Trivial. □

Theorem (Chain Rule)

Let $I, J \subset \mathbb{R}$ be intervals, $g : I \to \mathbb{R}$, $f : J \to I$, and let $c \in J$, with f differentiable at c and g differentiable at $f(c)$. Then $h = g \circ f : J \to \mathbb{R}$ is differentiable at c, with

$$h'(c) = g'(f(c))\, f'(c).$$

Proof

Use Carathéodory's theorem. □

Theorem

If $f : I \to f(I)$ is strictly increasing, with $g : f(I) \to I$ the inverse, and f is differentiable at $c \in I$ with $f'(c) \neq 0$, then g is differentiable at $f(c)$, and

$$g'(f(c)) = \frac{1}{f'(c)}.$$

Proof

The inverse g is continuous at $f(c)$. For $y \in f(I) \setminus \{f(c)\}$, put $x = g(y)$. As $y \to f(c)$, continuity of g gives $x \to c$. Moreover,

$$\frac{g(y) - g(f(c))}{y - f(c)} = \frac{x - c}{f(x) - f(c)} = \left(\frac{f(x) - f(c)}{x - c}\right)^{-1} \longrightarrow \frac{1}{f'(c)}.$$

Thus g is differentiable at $f(c)$, with $g'(f(c)) = 1/f'(c)$. □

Example (Rational power rule)

For every $\alpha \in \mathbb{Q}_{>0}$, the function

$$f(x) = x^\alpha$$

is differentiable on $(0, \infty)$, with

$$f'(x) = \alpha x^{\alpha - 1}.$$

Proof

Write $\alpha = p/q$, with $p, q \in \mathbb{N}_{>0}$. The function $h(u) = u^q$ is strictly increasing on $(0, \infty)$, with $h'(u) = qu^{q-1} \neq 0$. Its inverse is therefore differentiable and

$$(h^{-1})'(v) = \frac{1}{q(h^{-1}(v))^{q-1}}.$$

Since $x^{p/q} = h^{-1}(x^p)$, the chain rule gives

$$\frac{d}{dx} x^{p/q} = \frac{px^{p-1}}{q(x^{p/q})^{q-1}} = \frac{p}{q} x^{\frac{p}{q} - 1} = \alpha x^{\alpha - 1}.$$

□

6.2 Mean Value Theorem

Definition

Let $f : (a, b) \to \mathbb{R}$ and $c \in (a, b)$. The point c is called a local maximum of f if there exists $\delta > 0$ such that

$$|x - c| < \delta \quad \Longrightarrow \quad f(x) \leq f(c).$$

Likewise, it is a local minimum point of f if there exists $\delta > 0$ such that

$$f(x) \geq f(c), \qquad \forall |x - c| < \delta.$$

Lemma

Suppose $f : I \to \mathbb{R}$ and $c \in \mathrm{Int}(I)$. If c is a local maximum or local minimum of f, and f is differentiable at c, then $f'(c) = 0$.

$$\mathrm{Int}(A) = (\overline{A^c})^c.$$

Proof

Suppose first that c is a local maximum. For $x < c$ sufficiently close to c, the numerator and denominator of the difference quotient are both nonpositive, whereas for $x > c$ the numerator is nonpositive and the denominator positive. Hence

$$\lim_{x \to c^-} \frac{f(x) - f(c)}{x - c} \geq 0, \qquad \lim_{x \to c^+} \frac{f(x) - f(c)}{x - c} \leq 0.$$

Since $f'(c)$ exists, this implies $f'(c) \geq 0$ and $f'(c) \leq 0$, hence $f'(c) = 0$. If c is a local minimum, apply the maximum case to $-f$. □

One-sided notation

Do not confuse one-sided limits of the derivative with one-sided difference quotients based at c. In general,

$$\lim_{x \to c^-} f'(x) \neq \lim_{x \to c^-} \frac{f(x) - f(c)}{x - c}, \qquad \lim_{x \to c^+} f'(x) \neq \lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}.$$

Example

The function $f(x) = |x|$ has unequal one-sided difference quotients at $c = 0$.

Theorem (Mean Value Theorem)

Let $f : [a, b] \to \mathbb{R}$ be continuous and differentiable over (a, b). Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof

Consider the function

$$g(x) := f(x) - \frac{f(b) - f(a)}{b - a} x.$$

It is trivial to verify that g is continuous on $[a,b]$ and differentiable on (a,b). Claim: there exists $c \in (a,b)$ such that $g'(c) = 0$. If g is a constant function, then it is any point in (a,b). Suppose it is not. Then g is continuous, so g attains a maximum and a minimum on $[a,b]$. Moreover, g is not constant, hence at least one of the minimum or maximum is not at the endpoints. That point $c \in (a,b)$ is also a local minimum or maximum, so $g'(c) = 0$. Now

$$g'(c) = f'(c) - \frac{f(b) - f(a)}{b - a}$$

by the definition of g. Therefore

$$f'(c) - \frac{f(b) - f(a)}{b - a} = 0 \quad \Longrightarrow \quad f'(c) = \frac{f(b) - f(a)}{b - a}.$$

□

Corollary

Suppose $f : (a,b) \to \mathbb{R}$ is differentiable on (a,b). Then:

$$\begin{aligned}
f' \geq 0 &\implies f \text{ is monotone increasing,} \\
f' \leq 0 &\implies f \text{ is monotone decreasing,} \\
f' = 0 &\implies f \text{ is constant,} \\
f' < 0 &\implies f \text{ is strictly decreasing,} \\
f' > 0 &\implies f \text{ is strictly increasing.}
\end{aligned}$$

Proof

Let $(x_1, x_2) \subseteq (a,b)$. Then there exists $c \in (x_1, x_2)$ such that

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1)$$

by the Mean Value Theorem, which proves monotonicity. □

Theorem (Bounded derivative implies uniform continuity)

Let $f : I \to \mathbb{R}$, where $I \subset \mathbb{R}$ is a closed interval and f' exists on I. If f' is bounded, then f is uniformly continuous.

Proof

Choose $M > 0$ such that $|f'(t)| \leq M$ on I. For distinct $x, y \in I$, the Mean Value Theorem gives a point c between x and y such that

$$\frac{|f(x) - f(y)|}{|x - y|} = |f'(c)| \leq M.$$

Thus f is Lipschitz, and therefore uniformly continuous. □

Example (Uniform approximation by difference quotients)

Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable. If f' is continuous on $[a, b]$, where $a < b$, then for every $\varepsilon > 0$ there exists $\delta > 0$ such that, for every $x, y \in [a, b]$,

$$0 < |x - y| < \delta \quad \Longrightarrow \quad \left| \frac{f(x) - f(y)}{x - y} - f'(x) \right| < \varepsilon.$$

Proof

By the Heine–Cantor theorem, f' is uniformly continuous on $[a, b]$. Choose $\delta > 0$ such that

$$|u - v| < \delta \quad \Longrightarrow \quad |f'(u) - f'(v)| < \varepsilon$$

for all $u, v \in [a, b]$. If $0 < |x - y| < \delta$, the Mean Value Theorem gives a point c between x and y such that

$$\frac{f(x) - f(y)}{x - y} = f'(c).$$

Since $|c - x| < \delta$, the desired inequality follows. □

Theorem (Generalized Mean Value Theorem)

If $f, g : [a, b] \to \mathbb{R}$ are continuous and differentiable over (a, b),
then there exists $c \in (a, b)$ such that

$$f'(c)[g(b) - g(a)] = g'(c)[f(b) - f(a)].$$

Proof

Consider

$$h(x) := f(x)[g(b) - g(a)] - g(x)[f(b) - f(a)].$$

The function h is continuous on $[a, b]$, differentiable on (a, b), and $h(a) = h(b)$. Therefore there exists $c \in (a, b)$ such that $h'(c) = 0$, and hence

$$f'(c)[g(b) - g(a)] = g'(c)[f(b) - f(a)]. \qquad \square$$

Proposition (Darboux's theorem)

Suppose $f : [a, b] \to \mathbb{R}$ is differentiable at every point. For every $x_1, x_2 \in (a, b)$, with $x_1 < x_2$ and $f'(x_1) \neq f'(x_2)$, and every

$$\min\{f'(x_1), f'(x_2)\} < \lambda < \max\{f'(x_1), f'(x_2)\},$$

$$\exists x \in (x_1, x_2) \quad \text{such that} \quad f'(x) = \lambda.$$

This is the "intermediate value property for derivatives."

Proof

Without loss of generality, suppose $f'(x_1) < f'(x_2)$. Let $\lambda \in (f'(x_1), f'(x_2))$. Consider

$$g(x) := f(x) - \lambda x.$$

Since f is differentiable on $[a, b]$, g is differentiable on $[a, b]$. Then

$$g'(x_1) = f'(x_1) - \lambda < 0, \qquad g'(x_2) = f'(x_2) - \lambda > 0.$$

Moreover, since $g'(x_1) < 0$ and $g'(x_2) > 0$,

$$\begin{aligned} &\exists \delta_1 > 0 \text{ such that } \forall x_1 < x < x_1 + \delta_1, \quad g(x) < g(x_1), \\ &\exists \delta_2 > 0 \text{ such that } \forall x_2 - \delta_2 < x < x_2, \quad g(x) < g(x_2). \qquad (*) \end{aligned}$$

Since $[x_1, x_2]$ is compact, g attains the minimum value at a point $x \in [x_1, x_2]$ by the extreme value theorem. Moreover, $x \neq x_1, x_2$ by (*). Therefore $x \in (x_1, x_2)$, so x attains a local minimum and

$$g'(x) = 0 = f'(x) - \lambda \quad \Longrightarrow \quad f'(x) = \lambda, \qquad x \in (x_1, x_2).$$

□

Proposition

All functions on an interval of $\mathbb{R}$ that have the intermediate value property (as in Darboux's theorem) cannot have a discontinuity of first kind.

Proof

We prove that every function which has a discontinuity of first kind must not satisfy the intermediate value property. Let $f : (a, b) \to \mathbb{R}$ have a discontinuity of first kind at $x_0 \in (a, b)$. Then we must have either

$$f(x_0-) \neq f(x_0) \qquad \text{or} \qquad f(x_0+) \neq f(x_0).$$

Without loss of generality, suppose $f(x_0+) \neq f(x_0)$ and $f(x_0+) > f(x_0)$. Suppose

$$f(x_0+) - f(x_0) = \varepsilon > 0.$$

Then, by the definition of the limit, there exists $\delta > 0$ such that

$$0 < x - x_0 < \delta \quad \Longrightarrow \quad |f(x_0+) - f(x)| < \frac{\varepsilon}{2} \quad \Longrightarrow \quad f(x_0+) - \frac{\varepsilon}{2} < f(x) < f(x_0+) + \frac{\varepsilon}{2}.$$

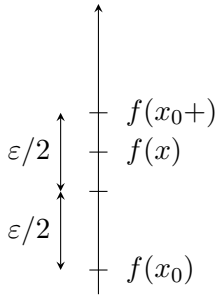
Choose $x \in (x_0, x_0 + \delta) \cap (a, b)$.
Then

$$f(x_0+) - \frac{\varepsilon}{2} < f(x) < f(x_0+) + \frac{\varepsilon}{2}.$$

We claim the intermediate value property fails on (x_0, x). Choose

$$f(x_0) + \frac{\varepsilon}{2} = f(x_0+) - \frac{\varepsilon}{2} < f(x).$$

The associated value-line diagram is:



Then there does not exist $t \in (x_0, x)$ such that $f(t) = f(x_0) + \varepsilon/2$, since $t \in (x_0, x)$ implies

$$0 < t - x_0 < x - x_0 < \delta \quad \Longrightarrow \quad f(t) > f(x_0) + \frac{\varepsilon}{2}.$$

Thus the intermediate value property fails. □

Corollary

If f is differentiable on (a, b), then f' cannot have a discontinuity of first kind.

Proof

By Darboux's theorem, $f' : (a, b) \to \mathbb{R}$ has the intermediate value property; then apply the previous proposition. □

6.3 L'Hôpital's Rule

Theorem (L'Hôpital)

Let

$$-\infty \leq a < b \leq \infty, \qquad f, g : \mathbb{R} \to \mathbb{R}$$

be differentiable on (a, b), with $g'(x) \neq 0$ for every $x \in (a, b)$. Suppose that

$$\lim_{x \to a^+} f(x) = \lim_{x \to a^+} g(x) = 0, \qquad \text{or} \qquad \lim_{x \to a^+} g(x) = \pm\infty.$$

Then, if

$$\lim_{x \to a^+} \frac{f'(x)}{g'(x)} = L \in \mathbb{R} \cup \{-\infty, \infty\},$$

we have

$$\lim_{x \to a^+} \frac{f(x)}{g(x)} = L.$$

Proof

Case 1.1: $L \in \mathbb{R}$ and $\lim_{x \to a^+} f(x) = \lim_{x \to a^+} g(x) = 0$. Let $\varepsilon > 0$. Since

$$\lim_{x \to a^+} \frac{f'(x)}{g'(x)} = L,$$

there exists $c \in (a, b)$ such that

$$a < x < c \quad \Longrightarrow \quad \left| \frac{f'(x)}{g'(x)} - L \right| < \varepsilon.$$

Let α, β satisfy $a < \alpha < \beta < c$. By the Generalized Mean Value Theorem, there exists $u \in (\alpha, \beta)$ such that

$$\frac{f'(u)}{g'(u)} = \frac{f(\beta) - f(\alpha)}{g(\beta) - g(\alpha)} \quad \Longrightarrow \quad \left| \frac{f(\beta) - f(\alpha)}{g(\beta) - g(\alpha)} - L \right| < \varepsilon.$$

Send $\alpha \to a^+$. We have

$$\left| \frac{f(\beta)}{g(\beta)} - L \right| \leq \varepsilon, \qquad \forall a < \beta < c.$$

We have shown that, for every $\varepsilon > 0$, there exists $c \in (a, b)$ such that, for every $a < \beta < c$,

$$\left| \frac{f(\beta)}{g(\beta)} - L \right| < \varepsilon.$$

Case 1.2: $L \in \{\pm\infty\}$ and $\lim_{x \to a^+} f(x) = \lim_{x \to a^+} g(x) = 0$. We do one case; the other one follows similarly. Suppose $L = \infty$. Let $N \in \mathbb{N}$. Then there exists $c \in (a, b)$ such that

$$a < x < c \quad \Longrightarrow \quad \frac{f'(x)}{g'(x)} \geq N.$$

Let $(\alpha, \beta) \subset (a, c)$. By the Generalized Mean Value Theorem again, there exists $u \in (\alpha, \beta) \subset (a, c)$ such that

$$\frac{f(\beta) - f(\alpha)}{g(\beta) - g(\alpha)} = \frac{f'(u)}{g'(u)} \geq N.$$

Sending $\alpha \to a^+$,

$$\frac{f(\beta)}{g(\beta)} \geq N.$$

Thus, for every $N \in \mathbb{N}$, there exists $c \in (a, b)$ such that

$$a < \beta < c \quad \Longrightarrow \quad \frac{f(\beta)}{g(\beta)} \geq N.$$

Therefore

$$\frac{f(x)}{g(x)} \xrightarrow[x \to a^+]{} \infty.$$

Case 2.1: $L \in \mathbb{R}$ and $\lim_{x \to a^+} g(x) = \pm\infty$. We want to show

$$\lim_{x \to a^+} \frac{f(x)}{g(x)} = L.$$

Let $\varepsilon > 0$. Since $\lim_{x \to a^+} f'(x)/g'(x) = L$, there exists $c \in (a, b)$ such that

$$a < x < c \quad \Longrightarrow \quad \left| \frac{f'(x)}{g'(x)} - L \right| < \varepsilon,$$

and hence

$$L - \varepsilon < \frac{f'(x)}{g'(x)} < L + \varepsilon, \qquad \forall x \in (a, c).$$

Let $\beta \in (a, c)$. By the Generalized Mean Value Theorem, there exists $u \in (\beta, c)$ such that

$$\frac{f'(u)}{g'(u)} = \frac{f(c) - f(\beta)}{g(c) - g(\beta)},$$

so

$$L - \varepsilon < \frac{f(\beta) - f(c)}{g(\beta) - g(c)} < L + \varepsilon.$$

Because $g(\beta) \to \pm\infty$, after decreasing c if necessary we may assume

$$1 - \frac{g(c)}{g(\beta)} > 0, \qquad a < \beta < c.$$

Therefore, for every $\beta \in (a, c)$,

$$(L - \varepsilon)\frac{g(\beta) - g(c)}{g(\beta)} < \frac{f(\beta)}{g(\beta)} - \frac{f(c)}{g(\beta)} < (L + \varepsilon)\frac{g(\beta) - g(c)}{g(\beta)},$$

and thus

$$L - \varepsilon + \frac{f(c) - Lg(c) + \varepsilon g(c)}{g(\beta)} < \frac{f(\beta)}{g(\beta)} < L + \varepsilon + \frac{f(c) - Lg(c) - \varepsilon g(c)}{g(\beta)}.$$

Since $\varepsilon g(c)$, $f(c)$, and $Lg(c)$ are all constants, and $g(\beta) \xrightarrow[\beta \to a^+]{} \pm\infty$, we can choose $\beta \in (a, c)$ close enough to a^+ so that the terms

$$\frac{\varepsilon g(c) + f(c) - Lg(c)}{g(\beta)}, \qquad \frac{f(c) - Lg(c) - \varepsilon g(c)}{g(\beta)}$$

are arbitrarily close to 0. Therefore

$$L - \varepsilon + \underbrace{\frac{f(c) - Lg(c) + \varepsilon g(c)}{g(\beta)}}_{\longrightarrow 0} < \frac{f(\beta)}{g(\beta)} < L + \varepsilon + \underbrace{\frac{f(c) - Lg(c) - \varepsilon g(c)}{g(\beta)}}_{\longrightarrow 0},$$

for β close enough to a^+. Hence

$$\frac{f(\beta)}{g(\beta)} \xrightarrow[\beta \to a^+]{} L.$$

Case 2.2: $L \in \{\pm\infty\}$ and $\lim_{x \to a^+} g(x) = \pm\infty$. We show the case with $L = +\infty$ ($L = -\infty$ likewise). We want to show

$$\lim_{x \to a^+} \frac{f(x)}{g(x)} = +\infty.$$

Let $N \in \mathbb{N}$. Then there exists $c \in (a, b)$ such that

$$a < x < c \quad \Longrightarrow \quad \frac{f'(x)}{g'(x)} \geq N.$$

Let $\beta \in (a, c)$. By the Generalized Mean Value Theorem again, there exists $u \in (\beta, c)$ such that

$$\frac{f'(u)}{g'(u)} = \frac{f(\beta) - f(c)}{g(\beta) - g(c)} \geq N,$$

Since $g(\beta) \to \pm\infty$ as $\beta \to a^+$, restrict β so that

$$1 - \frac{g(c)}{g(\beta)} > 0.$$

Multiplying by this positive factor gives

$$\frac{f(\beta) - f(c)}{g(\beta) - g(c)} \frac{g(\beta) - g(c)}{g(\beta)} \geq N \left(1 - \frac{g(c)}{g(\beta)}\right).$$

For such β,

$$\frac{f(\beta)}{g(\beta)} \geq N - N \underbrace{\frac{g(c)}{g(\beta)}}_{\to 0} + \underbrace{\frac{f(c)}{g(\beta)}}_{\to 0}.$$

The two error terms tend to 0, so for each N we may choose a cutoff c_N such that their sum is at least -1. Thus

$$\forall N \in \mathbb{N},\ \exists c_N \in (a,b) \text{ such that } \forall a < \beta < c_N, \qquad \frac{f(\beta)}{g(\beta)} \geq N - 1.$$

Therefore

$$\frac{f(\beta)}{g(\beta)} \xrightarrow[\beta \to a^+]{} +\infty.$$

□

A compact formulation

The argument can be summarized by two one-sided bounds. If $-\infty \leq L < +\infty$, then, whether $g(x) \to 0$ or $g(x) \to \pm\infty$,

$$\forall p > L,\ \exists c_1 \in (a,b) \text{ such that } a < x < c_1 \quad \Longrightarrow \quad \frac{f(x)}{g(x)} < p. \tag{*}$$

Similarly, if $-\infty < L \leq +\infty$, then, whether $g(x) \to 0$ or $g(x) \to \pm\infty$,

$$\forall q < L,\ \exists c_2 \in (a,b) \text{ such that } a < x < c_2 \quad \Longrightarrow \quad q < \frac{f(x)}{g(x)}. \tag{**}$$

For finite L, the two bounds together give the desired limit. For $L = +\infty$ or $L = -\infty$, the relevant single bound suffices.

6.4 Higher-Order Derivatives and Taylor's Theorem

Definition

Let $f : (a,b) \to \mathbb{R}$ be differentiable. If $f' : (a,b) \to \mathbb{R}$ is also differentiable, denote the derivative of f' by f'', or $f^{(2)}$, and call it the second-order derivative. Similarly,

$$f^{(k)}(x) = (f^{(k-1)})'(x), \qquad \forall x \in (a,b), \quad k \geq 1.$$

Theorem (Generalized L'Hôpital's theorem)

Suppose $f, g : (a, b) \to \mathbb{R}$ are n-times differentiable,

$$g^{(k)}(x) \neq 0, \qquad \forall k = 1, 2, \ldots, n, \quad \forall x \in (a, b), \qquad -\infty < a < b < +\infty.$$

Suppose

$$\lim_{x \to a^+} \frac{f^{(n)}(x)}{g^{(n)}(x)} = A,$$

and assume either that, for every $0 \leq k \leq n-1$,

$$\lim_{x \to a^+} f^{(k)}(x) = \lim_{x \to a^+} g^{(k)}(x) = 0,$$

or that

$$\lim_{x \to a^+} g(x) = \pm\infty.$$

Then

$$\lim_{x \to a^+} \frac{f(x)}{g(x)} = A.$$

Proof

Proceed by induction on n. In the zero-over-zero case, the hypotheses for the intermediate derivative ratios are part of the assumptions. In the infinite-denominator case, the nonvanishing of successive derivatives implies that each $g^{(k)}$ is strictly monotone. Since the endpoint a is finite, $g^{(k-1)}(x) \to \pm\infty$ forces $g^{(k)}(x) \to \pm\infty$; otherwise integrating a bounded derivative over a finite interval could not produce divergence. Thus L'Hôpital's rule applies successively to $f^{(k)}/g^{(k)}$, for $k = n-1, n-2, \ldots, 0$, and yields the result. □

Theorem (Taylor's theorem)

Suppose $f : (a, b) \to \mathbb{R}$ is an $(n+1)^{\text{th}}$-order differentiable function on (a, b). Then, for every $x, x_0 \in (a, b)$, there exists

$$c \in (x_0, x) \quad \{\text{or } (x, x_0)\}$$

such that

$$f(x) = \underbrace{\sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k}_{\substack{n^{\text{th}}\text{-order Taylor} \\ \text{polynomial of } f \text{ at } x_0}} + \underbrace{\frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}}_{n^{\text{th}}\text{-order remainder term}}. \qquad (*)$$

The same statement holds on a closed interval, with the usual one-sided interpretation at its endpoints. The displayed remainder is Lagrange's form.

Proof

Let $x, x_0 \in (a,b)$. If $x = x_0$, the identity is immediate. Suppose without loss of generality that $x_0 < x$; the case $x < x_0$ is analogous. Define

$$M_{x,x_0} := \frac{f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k}{(x-x_0)^{n+1}},$$

where x is treated as a constant. Define

$$g(s) := f(s) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(s-x_0)^k - M_{x,x_0}(s-x_0)^{n+1}.$$

Then

$$g(x) = f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k - M_{x,x_0}(x-x_0)^{n+1} = 0$$

by the definition of M_{x,x_0}, and

$$g(x_0) = f(x_0) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x_0-x_0)^k - M_{x,x_0}(x_0-x_0)^{n+1} = f(x_0) - \frac{f^{(0)}(x_0)}{1}(x_0-x_0)^0 - 0 = 0.$$

Similarly, we can check that

$$g^{(k)}(x_0) = 0, \qquad \forall 0 \le k \le n.$$

Since $g(x) = g(x_0) = 0$, repeated applications of the Mean Value Theorem give

$$\begin{aligned}
&\exists x_1 \in (x_0, x) && \text{such that } g'(x_1) = 0,\\
&\exists x_2 \in (x_0, x_1) && \text{such that } g''(x_2) = 0,\\
&&&\vdots\\
&\exists x_n \in (x_0, x_{n-1}) && \text{such that } g^{(n)}(x_n) = 0,\\
&\exists c \in (x_0, x_n) && \text{such that } g^{(n+1)}(c) = 0.
\end{aligned}$$

This c has the required property because

$$g^{(n+1)}(c) = f^{(n+1)}(c) - M_{x,x_0}(n+1)! = 0,$$

so

$$\frac{f^{(n+1)}(c)}{(n+1)!} = M_{x,x_0} = \frac{f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k}{(x-x_0)^{n+1}}.$$

Therefore

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x-x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-x_0)^{n+1}, \qquad c \in (x_0, x_n) \subset (x_0, x).$$

□

Theorem (Peano form of remainder)

Let $f : (a, b) \to \mathbb{R}$ be n^{th}-order differentiable over (a, b). Then, for every $x, x_0 \in (a, b)$,

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + R_n(x, x_0), \qquad \lim_{x \to x_0} \frac{R_n(x, x_0)}{(x - x_0)^n} = 0.$$

Proof

By definition,

$$f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k = R_n(x, x_0),$$

and both this expression and $(x - x_0)^n$ are n^{th}-order differentiable, with

$$\lim_{x \to x_0} \left[f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \right]^{(j)} = 0, \qquad \lim_{x \to x_0} \left[(x - x_0)^n \right]^{(j)} = 0,$$

for every $0 \le j \le n - 2$. Thus we can apply the generalized L'Hôpital's rule:

$$\begin{aligned} \lim_{x \to x_0} \frac{R_n(x, x_0)}{(x - x_0)^n} &= \lim_{x \to x_0} \frac{\left[f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k \right]^{(n-1)}}{\left[(x - x_0)^n \right]^{(n-1)}} \\ &= \lim_{x \to x_0} \frac{f^{(n-1)}(x) - f^{(n-1)}(x_0) - f^{(n)}(x_0)(x - x_0)}{n!(x - x_0)}. \qquad (*) \end{aligned}$$

Since $f^{(n)}(x_0)$ exists, set

$$\Delta(x) := f^{(n-1)}(x) - f^{(n-1)}(x_0).$$

Then

$$\lim_{x \to x_0} \frac{\Delta(x)}{x - x_0} - f^{(n)}(x_0) = 0,$$

and hence

$$\begin{aligned} 0 &= \lim_{x \to x_0} \left[\frac{\Delta(x)}{x - x_0} - f^{(n)}(x_0) \right] \\ &= \lim_{x \to x_0} \frac{\Delta(x) - f^{(n)}(x_0)(x - x_0)}{x - x_0}. \end{aligned}$$

Plugging back into (*),

$$\lim_{x \to x_0} \frac{R_n(x, x_0)}{(x - x_0)^n} = 0 \cdot \frac{1}{n!} = 0.$$

□

Corollary (Second derivative test)

Suppose $f : (a, b) \to \mathbb{R}$ is twice differentiable, with $f'(c) = 0$, $c \in (a, b)$. If $f''(c) > 0$, then c is a local minimum; if $f''(c) < 0$, then c is a local maximum.

Proof

Recall Taylor's theorem with the Peano form of remainder. Since f is twice differentiable,

$$f(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2}(x-c)^2 + R_2(x,c), \qquad \lim_{x\to c} \frac{R_2(x,c)}{(x-c)^2} = 0.$$

Thus, since $f'(c) = 0$,

$$f(x) - f(c) = (x-c)^2 \left(\frac{f''(c)}{2} + \frac{R_2(x,c)}{(x-c)^2} \right).$$

Moreover,

$$\lim_{x\to c} \frac{R_2(x,c)}{(x-c)^2} = 0 \quad \text{and} \quad f''(c) > 0$$

imply that there exists $\delta > 0$ such that

$$\frac{f''(c)}{2} + \frac{R_2(x,c)}{(x-c)^2} > 0.$$

Therefore, for every x with $0 < |x-c| < \delta$,

$$f(x) - f(c) = (x-c)^2(\cdots) > 0,$$

so $f(x) > f(c)$, and $f(c)$ is a local minimum. Local maximum likewise. □

Remark (Inconclusive case)

The test is inconclusive when $f''(c) = 0$. For example, at $c = 0$,

$$f(x) = x^3, \qquad f(x) = x^4.$$

7 Convexity

Definition

A function $f : (a,b) \to \mathbb{R}$ is convex on (a,b) if, for every $x_1, x_2 \in (a,b)$ and every $t \in [0,1]$,

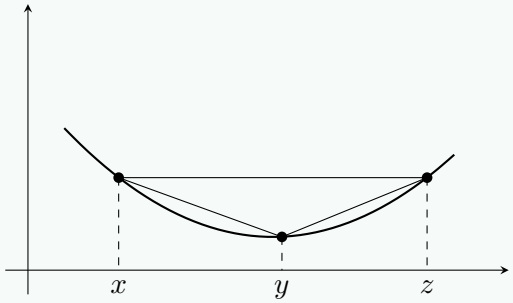
$$f\big((1-t)x_1 + tx_2\big) \le (1-t)f(x_1) + tf(x_2).$$

Proposition

If $f : (a,b) \to \mathbb{R}$ is convex, then, for every $a < x < y < z < b$,

$$\frac{f(y) - f(x)}{y - x} \le \frac{f(z) - f(x)}{z - x} \le \frac{f(z) - f(y)}{z - y}.$$

The three points on the convex curve and their corresponding secant lines are shown below:



Proof

We want to show

$$f(y) \leq f(x) + \frac{y-x}{z-x}(f(z) - f(x)) = \left(1 - \frac{y-x}{z-x}\right) f(x) + \frac{y-x}{z-x} f(z).$$

Notice

$$\left(1 - \frac{y-x}{z-x}\right) x + \frac{y-x}{z-x} z = \frac{y(z-x)}{z-x} = y.$$

Since $x < y < z$,

$$0 < \frac{y-x}{z-x} < 1.$$

Thus, by convexity,

$$f(y) = f\left(\left(1 - \frac{y-x}{z-x}\right) x + \frac{y-x}{z-x} z\right) \leq \left(1 - \frac{y-x}{z-x}\right) f(x) + \frac{y-x}{z-x} f(z),$$

so

$$\frac{f(y) - f(x)}{y-x} \leq \frac{f(z) - f(x)}{z-x}.$$

Similarly, noticing that

$$y = \left(1 - \frac{z-y}{z-x}\right) z + \frac{z-y}{z-x} x,$$

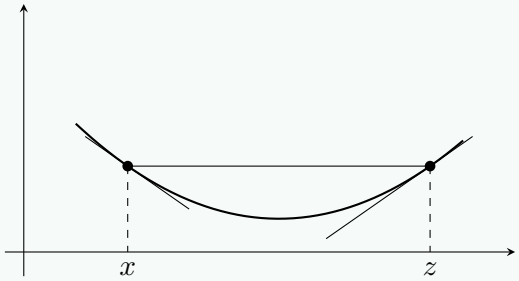
along with $0 < \frac{z-y}{z-x} < 1$, convexity proves the second half of the inequality. □

Proposition

If $f : (a, b) \to \mathbb{R}$ is a convex function and differentiable, then

$$f'(x) \leq f'(z), \qquad \forall a < x < z < b.$$

The diagram shows a convex graph, two tangent lines, and the secant through x and z:



Proof

By the preceding secant-slope proposition, for every $y \in (x, z)$,

$$\frac{f(y) - f(x)}{y - x} \leq \frac{f(z) - f(x)}{z - x} \leq \frac{f(z) - f(y)}{z - y}.$$

By differentiability, we can send $y \to x^+$ and $y \to z^-$. Since limits preserve loose inequality,

$$f'(x) \leq \frac{f(z) - f(x)}{z - x} \leq f'(z).$$

□

Theorem

If $f : (a, b) \to \mathbb{R}$ is twice differentiable, then

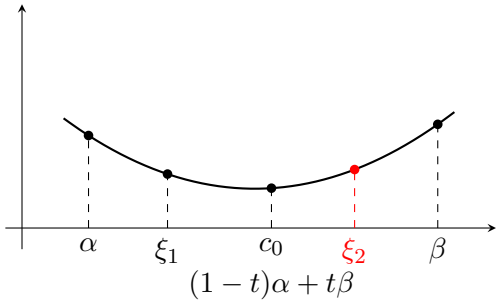
$$f \text{ is convex} \iff f''(x) \geq 0, \qquad \forall x \in (a, b).$$

Proof

($\Rightarrow$) Suppose f is convex and $f''(x) < 0$. Then there exists $h > 0$ such that

$$\frac{f'(x + h) - f'(x)}{h} < 0 \implies f'(x + h) < f'(x).$$

But $x + h > x$, contradicting the last proposition.
($\Leftarrow$) Suppose $f'' \geq 0$ on (a, b). The points $\alpha < \xi_1 < c_0 < \xi_2 < \beta$, with $c_0 = (1 - t)\alpha + t\beta$, are arranged as follows:



Let $\alpha, \beta \in (a, b)$, $\alpha < \beta$. Let $0 < t < 1$, and

$$c_0 := (1 - t)\alpha + t\beta.$$

By Taylor's theorem with Lagrange's form of remainder, there exist $\xi_1 \in (\alpha, c_0)$, $\xi_2 \in (c_0, \beta)$, such that

$$f(\beta) = f(c_0) + f'(c_0)(\beta - c_0) + \frac{f''(\xi_2)}{2}(\beta - c_0)^2,$$

and

$$f(\alpha) = f(c_0) + f'(c_0)(\alpha - c_0) + \frac{f''(\xi_1)}{2}(\alpha - c_0)^2.$$

Therefore

$$\begin{aligned}(1-t)f(\alpha)+tf(\beta) &= f(c_0)+\big[(1-t)(\alpha-c_0)+t(\beta-c_0)\big]f'(c_0)\\ &\quad+\underbrace{\frac{f''(\xi_2)}{2}(\beta-c_0)^2t+\frac{f''(\xi_1)}{2}(\alpha-c_0)^2(1-t)}_{\geq 0}\\ &\geq f(c_0)+[\alpha-c_0+t\beta-t\alpha]f'(c_0)\\ &= f(c_0)+\big[\alpha-((1-t)\alpha+t\beta)+t(\beta-\alpha)\big]f'(c_0)\\ &= f(c_0)+0=f(c_0)=f((1-t)\alpha+t\beta).\end{aligned}$$

Thus f is convex. □